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WEARABLE DEVICE FOR SWITCHING AT LEAST ONE APPLICATION ON BASIS OF TYPE OF EXTERNAL ELECTRONIC DEVICE AND METHOD THEREFOR

Abstract

A processor of a wearable device according to an embodiment may display at least one first screen on a display. In response to identifying a change in the state of an external electronic device to a first state, the processor may transmit information about at least one first screen displayed on the display of the wearable device to the external electronic device in order to display at least one second screen related to the at least one first screen via the external electronic device. In response to identifying a change in the state of the external electronic device from the first state to a second state different, the processor may receive information about at least one third screen displayed via the external electronic device. The processor may display, on the display of the wearable device, at least one fourth screen related to the at least one second screen being displayed via the display of the external electronic device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of PCT International Application No. PCT/KR2023/021954, which was filed on Dec. 28, 2023, and claims priority to Korean Patent Application No. 10-2023-0009746, filed on Jan. 25, 2023, in the Korean Intellectual Property Office, and claims priority to Korean Patent Application No. 10-2023-0001407, filed on Jan. 4, 2023, in the Korean Intellectual Property Office, the disclosure of which are incorporated by reference herein their entirety.

BACKGROUND
Technical Field
[0002] The present disclosure relates to a wearable device for changing at least one application based on a form of an external electronic device and a method thereof.
Description of Related Art
[0003] In order to provide an enhanced user experience, an electronic device providing an augmented reality (AR) service, which displays information generated by computer in association with an external object within the real-world, are being developed. The electronic device may be a wearable device that may be worn by a user. For example, the electronic device may be an AR glasses and/or a head-mounted device (HMD).
SUMMARY
[0004] According to an embodiment, a wearable device may comprise a communication circuitry, a display, memory configured to store instructions, and a processor. The instructions, when executed by the processor, cause the wearable device to display within the display at least one first screen. The instructions, when executed by the processor, cause the wearable device to transmit, in response to identifying that an external electronic device is transformed into a first state based on the information, to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device. The instructions, when executed by the processor, cause the wearable device to transmit, in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, information associated with at least one third screen displayed through the external electronic device. The instructions, when executed by the processor, cause the wearable device to display within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device.
[0005] The non-transitory computer readable storage medium storing instructions may be provided.

The instructions, when executed by an wearable device including a communication circuitry, and a display, cause the wearable device to display within the display at least one first screen. The instructions, when executed by the wearable device, cause the wearable device to transmit, in response to identifying that an external electronic device is transformed into a first state based on the information, to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device. The instructions, when executed by the wearable device, cause the wearable device to transmit, in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, information associated with at least one third screen displayed through the external electronic device. The instructions, when executed by the wearable device, cause the wearable device to display within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device on the display of the wearable device.

[0006] According to an embodiment, a wearable device may comprise a communication circuitry, a display, memory configured to store instructions, and a processor. The instructions, when executed by the processor, may cause the wearable device to obtain information with respect to a form of an external electronic device. The instructions, when executed by the processor, may cause the wearable device to, in response to identifying that the external electronic device is transformed into a first form based on the information, execute a function to move at least one first screen being displayed on the display of the wearable device to a display of the external electronic device. The instructions, when executed by the processor, may cause the wearable device to, in response to identifying that the external electronic device is transformed into a second form different from the first form based on the information, display at least one second screen being displayed through the display of the external electronic device on the display of the wearable device.

[0007] According to an embodiment, a method performed by a wearable device may comprise obtaining information with respect to a form of an external electronic device. The method may comprise, in response to identifying that the external electronic device is transformed into a first form based on the information, executing a function to move at least one first screen being displayed in a display of the wearable device to a display of the external electronic device. The method may comprise, in response to identifying that the external electronic device is transformed into a second form different from the first form based on the information, displaying at least one second screen being displayed through the display of the external electronic device on the display of the wearable device.

[0008] According to an embodiment, a wearable device may comprise a camera, a communication circuit, a display, memory storing instructions, and a processor. The instructions, when executed by the processor, may cause the wearable device to obtain information on a form of the external electronic device, in a state in which an external electronic device viewed through the display by using the camera is identified. The instructions, when executed by the processor, may cause the wearable device to display a visual object on the display to guide the movement of at least one media content, based on identifying transformation of a flexible display of the external electronic device by using the information. The instructions, when executed by the processor, may cause the wearable device to, while the visual object is displayed, execute a function for moving at least one media content between the wearable device and the external electronic device, based on transformation of the flexible display viewed through the display.

[0009] According to an embodiment, a method of a wearable device may comprise obtaining information on a form of the external electronic device, in a state in which an external electronic device viewed through a display of the wearable device by using a camera of the wearable device is identified. The method may comprise displaying a visual object on the display to guide the movement of at least one media content, based on identifying transformation of a flexible display of the external electronic device by using the information. The method may comprise, while the visual object is displayed, executing a function for moving at least one media content between the wearable device and the external electronic device, based on transformation of the flexible display viewed through the display.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** illustrates an example of an operation in which a wearable device moves a screen between an external electronic device and the wearable device, according to an embodiment.

[0011] FIG. **2** illustrates an example of a block diagram of a wearable device, according to an embodiment.

[0012] FIG. **3** illustrates an example of a flowchart of operations performed by a wearable device, according to an embodiment.

[0013] FIG. **4** illustrates an example of an operation in which a wearable device moves media content to an external electronic device, according to an embodiment.

[0014] FIG. **5** illustrates an example of an operation in which a wearable device moves media content displayed through an external electronic device to the wearable device, according to an embodiment.

[0015] FIG. **6** illustrates an example of a flowchart of operations performed by a wearable device, according to an embodiment.

[0016] FIGS. **7**A to **7**B illustrate an example of a signal flowchart between a wearable device and an external electronic device, according to an embodiment.

[0017] FIGS. **8**A to **8**B illustrate an example of an operation performed by a wearable device based on a form of an external electronic device, according to an embodiment.

[0018] FIG. **9** illustrates an example of a flowchart of operations performed by a wearable device, according to an embodiment.

[0019] FIGS. **10**A to **10**B illustrate an example of a signal flowchart between a wearable device and an external electronic device, according to an embodiment.

[0020] FIGS. **11**A, **11**B, and **11**C illustrate an example of an operation in which a wearable device moves at least one application based on a form of an external electronic device, according to an embodiment.

[0021] FIG. **12** illustrates an example of an operation performed by a wearable device based on a direction of an external electronic device, according to an embodiment.

[0022] FIG. **13** illustrates an example of an operation performed by a wearable device based on a form of an external electronic device, according to an embodiment.

[0023] FIG. **14**A illustrates an example of a perspective view of a wearable device, according to an embodiment.

[0024] FIG. **14**B illustrates an example of one or more hardware disposed in a wearable device, according to an embodiment.

[0025] FIGS. **15**A to **15**B illustrate an example of an exterior of a wearable device, according to an embodiment.

DETAILED DESCRIPTION

[0026] Hereinafter, various embodiments of the present document will be described with reference to the accompanying drawings.

[0027] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0028] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0029] FIG. **1** illustrates an example of an operation in which a wearable device **101** moves a screen between an external electronic device **120** and the wearable device **101**, according to an embodiment. In an embodiment, the wearable device **101** may include a head-mounted display (HMD) wearable on a head of a user **110**. Although an exterior of the wearable device **101** having a form of glasses is illustrated, embodiment is not limited thereto. An example of one or more hardware included in the wearable device **101** will be described with reference to FIG. **2**. An example of a structure of the wearable device **101** wearable on a head of the user **110** will be described with reference to FIGS. **14**A to **14**B and/or **15**A to **15**B. The wearable device **101** may be referred to as an electronic device. For example, the electronic device may be combined with an accessory for attachment to the user's head to form the HMD.

[0030] According to an embodiment, the wearable device **101** may execute a function related to video see-through (VST) and/or virtual reality (VR). Referring to FIG. **1**, in a state in which the user **110** wears the wearable device **101**, the wearable device **101** may include a housing covering the user **110**'s eyes. The wearable device **101** may include a display disposed on a first surface of the housing facing the eyes in the state. The wearable device **101** may include a camera disposed on a second surface opposite to the first surface. The wearable device **101** may obtain frames including ambient light, by using the camera. The wearable device **101** may output the frames in a display disposed on the first surface so that the user **110** may recognize the ambient light through the display. A displaying region of the display disposed on the first surface may be formed by one or more pixels included in the display. The wearable device **101** may synthesize a virtual object in frames outputted through the display so that the user **110** may recognize the virtual object together with a real object recognized by the ambient light.

[0031] According to an embodiment, the wearable device **101** may execute a function related to augmented reality (AR) and/or mixed reality (MR). As shown in FIG. **1**, in a state in which the user **110** wears the wearable device **101**, the wearable device **101** may include at least one lens disposed adjacent to the user **110**'s eyes. The wearable device **101** may combine light emitted from the display of the wearable device **101** with ambient light passing through the lens. The displaying region of the display may be formed within a lens through which ambient light passes. Since the wearable device **101** combines the ambient light and the light emitted from the display, the user **110** may see an image that is a mixture of a real object recognized by the ambient light and a virtual object formed by the light emitted from the display.

[0032] In an embodiment, the wearable device **101** may communicate with the external electronic device **120** in order to support controllability of a plurality of electronic devices including the wearable device **101**. Referring to FIG. **1**, exemplary states **191** and **192** in which the wearable device **101** is connected to the external electronic device **120** are illustrated. The external electronic device **120** may be a terminal able to be owned by the user **110**. Referring to FIG. **1**, the external electronic device **120** including a housing foldable by a folding axis F is exemplarily illustrated. According to an embodiment, the wearable device **101** may perform moving (or exchanging or transferring) of information between the external electronic device **120** and the wearable device **101**, based on a form of the external electronic device **120** that is deformed with respect to the folding axis F. The information moved between the external electronic device **120** and the wearable device **101** may include an image, video, text, or a combination thereof that may be outputted through a display of the wearable device **101** and/or the external electronic device **120**. The information moved between the external electronic device **120** and the wearable device **101** may include a screen provided from at least one application. The screen may be displayed based on a region assigned to the application, such as activity and/or a window. The screen, as an execution screen of an application, may include, which is generated based on execution of the application, text, an image, a video, a UI for interacting with a user, or any combination thereof

[0033] Referring to FIG. **1**, in state **191**, according to an embodiment, the wearable device **101** may provide a user experience based on VST through a display covering two eyes of the user **110**. For example, the wearable device **101** may display at least one virtual object together with an image of at least a portion of external environment (e.g., a portion of external environment in which the user **110**′s eyes faces). Referring to the exemplary state **191** of FIG. **1**, the at least one virtual object displayed by the wearable device **101** may include icons **131** representing an application. The at least one virtual object may include screens **132**, **133**, **134**, and **135** provided from at least one application executed by the wearable device **101**. Based on the execution of the application, the wearable device **101** may display a widget (e.g., the screen **133** having a watch form) and/or a window (e.g., the screen **132** provided by an application to display a schedule, the screen **134** provided by an application to display weather information, and/or the screen **135** provided by an application for video conference). The at least one virtual object may include a screen **140** displayed for reproducing media content including video. The wearable device **101** may display a visual object, such as a play button **141**, for controlling reproduction of media content corresponding to the screen **140** on the screen **140**.

[0034] According to an embodiment, the wearable device **101** may execute a function for moving the screen displayed through the display of the wearable device **101** to an external electronic device **120** different from the wearable device **101**. The wearable device **101** may execute the function based on recognizing or identifying a change of the form of the external electronic device **120**. Deformation of the external electronic device **120** may cause deformation of a flexible display included in the external electronic device **120**. The term “deformation” can be understood as a change, transition or transformation of a form of the external electronic device **120** operated by a user. According to an embodiment, an operation in which the wearable device **101** identifies the deformation of the flexible display and executes the function based on the identified deformation will be described with reference to FIG. **3**. As an example of an external electronic device **120** including a deformable flexible display, an external electronic device **120** including a housing foldable by a folding axis F is exemplarily illustrated. An operation performed by the wearable device **101** to exchange information with the external electronic device **120** including a straight folding axis F will be described with reference to FIGS. **4** to **12**. The embodiment is not limited thereto, and the wearable device **101** may exchange information with an external electronic device **120** that includes a flexible display insertable into a housing based on a rollable structure. In response to deformation of the external electronic device **120** having the rollable structure, an operation performed by the wearable device **101** will be described with reference to FIG. **13**.

[0035] In the state **191** of FIG. **1**, the wearable device **101** may obtain information on a form (or a state) of the external electronic device **120** adjacent to the wearable device **101**. In the exemplary state **191** in which the wearable device **101** is connected to the external electronic device **120** including a housing foldable by folding axis F, the information may include an angle of folding axis F. For example, the form of the external electronic device **120** may be parameterized by the angle of folding axis F. Referring to FIG. **1**, the user **110** may unfold the flexible display included in the external electronic device **120** by transforming the external electronic device **120**. While the user **110** unfolds the external electronic device **120**, the wearable device **101** may identify the unfolding of the flexible display facing the user **110**. For example, when the angle of folding axis F is included in a preset angle range including 180°, the external electronic device **120** and/or the flexible display may be fully unfolded. Based on identifying the fully unfolded external electronic device **120**, the wearable device **101** may move at least one virtual object displayed through a display of the wearable device **101** to the external electronic device **120**.

[0036] Referring to FIG. **1**, as the external electronic device **120** is changed from the state **191** to a fully unfolded state **192**, the wearable device **101** may move media content and/or screen displayed in the state **191** to the external electronic device **120**. Referring to FIG. **1**, the exemplary state **192** in which the wearable device **101** moves the screen **140** in the state **191** to the external electronic device **120** is illustrated. In the state **192**, as the screen **140** is moved from the wearable device **101** to the external electronic device **120**, media content **160** (e.g., video) that was reproduced through the screen **140** may be displayed on a displaying region **150** formed by the flexible display of the external electronic device **120**.

[0037] In an embodiment, the wearable device **101** may transmit information (e.g., media content **160**) related to the screen **140** to the external electronic device **120**, in order to continuously move the screen **140** from the display of the wearable device **101** to the flexible display of the external electronic device **120**. The wearable device **101** may transmit a signal including the information to the external electronic device **120**, based on a form of the external electronic device **120**. The wearable device **101** may move the screen **140** to the external electronic device **120** based on a motion of the user **110** as well as the form of the external electronic device **120**.

[0038] In an embodiment, the motion of the user **110** identified by the wearable device **101** for moving the screen **140** may include a gaze of the user **110** facing a specific virtual object (e.g., the screen **140**) and/or a motion of the user **110** moving a specific virtual object into a preset region of the display of the wearable device **101**. The wearable device **101** may include a camera disposed toward the user **110**'s eyes to identify the gaze of the user **110**. The wearable device **101** may identify a direction of the gaze by using a form of eyes included in an image of the camera. In the state **191** in which the gaze faces the screen **140**, in response to identifying that the external electronic device **120** is fully unfolded, the wearable device **101** may change to the state **192** in which the media content **160** is displayed through the external electronic device **120**. An operation in which the wearable device **101** exchanges information between the wearable device **101** and the external electronic device **120** by using the user's gaze will be described with reference to FIGS. **4** to **6**, **7**A to **7**B. An operation in which the wearable device **101** exchanges the information based on a position of the external electronic device **120** will be described with reference to FIGS. **8**A to **8**B, **9**, and **10**A to **10**B.

[0039] As described above, as the external electronic device **120** is unfolded, a state of the wearable device **101** may be changed from the state **191** to the state **192**. The embodiment is not limited thereto, and as the external electronic device **120** is folded, the wearable device **101** may move at least one screen displayed through the flexible display of the external electronic device **120** to the display of the wearable device **101**. In the state **192** of identifying the fully unfolded external electronic device **120**, the wearable device **101** may identify changing of a form of the external electronic device **120**, based on information on the form of the external electronic device **120**. For example, the wearable device **101** may obtain information for displaying the media content **160** of the displaying region **150** on the display of the wearable device **101**, based on identifying that an angle of folding axis F is reduced to less than a threshold angle smaller than 180°. For example, the wearable device **101** may request information related to at least one screen displayed through the displaying region **150** from the external electronic device **120**, based on identifying the folding of the external electronic device **120**. In response to identifying that the external electronic device **120** is fully folded from the state **192** in which the external electronic device **120** is fully unfolded, the wearable device **101** may display a screen (e.g., the screen **140** of the state **191**) for reproducing the media content **160** using the wearable device **101**, based on the information. The screen displayed by the wearable device **101** may be displayed in a virtual space and/or a virtual region provided by the wearable device **101**.

[0040] As described above, according to an embodiment, the wearable device **101** may control movement of information between the external electronic device **120** and the wearable device **101** according to the form of the external electronic device **120**. As the information is moved, at least one virtual object (e.g., the screen **140**) displayed in the wearable device **101** may be moved to the external electronic device **120**. For example, the at least one virtual object that was displayed by the wearable device **101** may be displayed through the external electronic device **120** from the moment when the external electronic device **120** is transformed into a preset form. Similarly, as the information is moved, the media content **160** displayed in the external electronic device **120** may be moved to the wearable device **101**. Since the information is moved according to deformation of the external electronic device **120**, the wearable device **101** may recognize the deformation of the external electronic device **120** as a gesture for moving the information.

[0041] As shown in FIG. **1**, in a state identifying of the external electronic device **120** including a flexible display foldable by a folding axis F, the wearable device **101** may compare a form of the external electronic device **120** with a preset form for moving the information, according to an angle by which the flexible display is folded with respect to the folding axis F. The preset form may include a first preset form (e.g., an unfolded form of the external electronic device **120**) for moving at least one screen from the wearable device **101** to the external electronic device **120**. The preset form may include a second preset form (e.g., a folded form of the external electronic device **120**) for moving at least one screen from the external electronic device **120** to the wearable device **101**.

[0042] Hereinafter, referring to FIG. **2**, according to an embodiment, an example of hardware and/or software that the wearable device **101** includes for movement of information between the external electronic device **120** and the wearable device **101** is described.

[0043] FIG. **2** illustrates an example of a block diagram of a wearable device **101**, according to an embodiment. The wearable device **101** of FIG. **2** may include the wearable device **101** of FIG. **1**. An external electronic device **120** of FIG. **2** may include the external electronic device **120** of FIG. **1**.

[0044] According to an embodiment, the wearable device **101** may include at least one of a processor **210-1**, a memory **215-1**, a display **220-1**, a camera **225**, a sensor **230-1**, or a communication circuit **240-1**. The processor **210-1**, the memory **215-1**, the display **220-1**, the camera **225**, the sensor **230-1**, and the communication circuit **240-1** may be electronically and/or operably coupled with each other by an electronical component such as a communication bus **202-1**. Hereinafter, operational coupling of hardware may mean that a direct or indirect connection between hardware is established by wire or wirelessly, so that a second hardware is controlled by a first hardware among the hardware. Although illustrated through different blocks, embodiments are not limited thereto, and a portion (e.g., at least a portion of the processor **210-1**, the memory **215-1** and the communication circuit **240-1**) of the hardware of FIG. **2** may be included in a single integrated circuit such as a system on a chip (SoC). The type and/or number of hardware included in the wearable device **101** is not limited to those illustrated in FIG. **2**. For example, the wearable device **101** may include only a portion of the hardware illustrated in FIG. **2**.

[0045] In an embodiment, the processor **210-1** of the wearable device **101** may include hardware for processing data according to one or more instructions. For example, the hardware for processing data may include an arithmetical and logical unit (ALU), a floating point unit (FPU), a field programmable gate array (FPGA), a central processing unit (CPU), and/or an application processor (AP). The processor **210** may have a structure of a single-core processor, or a structure of a multi-core processor such as a dual core, a quad core, or a hexa core.

[0046] According to an embodiment, the memory **215-1** of the wearable device **101** may include a hardware component for storing data and/or instructions inputted and/or outputted in the processor **210-1** of the wearable device **101**. The memory **215-1** may include a volatile memory such as a random-access memory (RAM) and/or a non-volatile memory such as a read-only memory (ROM). For example, the volatile memory may include at least one of a dynamic RAM (DRAM), a static RAM (SRAM), a Cache RAM, and a pseudo SRAM (PSRAM). For example, the non-volatile memory may include at least one of a programmable ROM (PROM), an erasable PROM (EPROM), an electrically erasable PROM (EEPROM), a flash memory, a hard disk, a compact disk, a solid state drive (SSD), and an embedded multimedia card (eMMC).

[0047] In an embodiment, the display **220-1** of the wearable device **101** may output visualized information (e.g., screens of FIGS. **1**, **4** to **5**, **8**A to **8**B, **11**A to **11**C, and/or **12** to **13**) to a user (e.g., the user **110** of FIG. **1**). For example, the display **220-1** may be controlled by the processor **210-1** including a circuit such as a graphic processing unit (GPU) to output visualized information to the user. The display **220-1** may include a flat panel display (FPD) and/or an electronic paper. The FPD may include a liquid crystal display (LCD), a plasma display panel (PDP), and/or one or more light emitting diode (LED). The LED may include an organic LED (OLED). The display **220-1** of FIG. **2** may include at least one display **1450** to be described later with reference to FIGS. **14**A to **14**B and/or **15**A to **15**B.

[0048] In an embodiment, the camera **225** of the wearable device **101** may include one or more optical sensors (e.g., a charged coupled device (CCD) sensor, a complementary metal oxide semiconductor (CMOS) sensor) that generate an electrical signal indicating color and/or brightness of light. A plurality of optical sensors included in the camera **225** may be arranged in a form of a 2-dimensional array. The camera **225** may generate 2-dimensional frame data corresponding to light reaching optical sensors of the 2-dimensional array by obtaining electrical signals of each of the plurality of optical sensors substantially simultaneously. For example, photographic data captured using the camera **225** may mean a two-dimensional frame data obtained from the camera **225**. For example, video data captured using the camera **225** may mean a sequence of a plurality of 2-dimensional frame data obtained from the camera **225** according to a frame rate. The camera **225** is disposed toward a direction in which the camera **225** receives light, and may further include a flashlight for outputting light in the direction.

[0049] Although the camera **225** is illustrated based on a single block, the number of camera **225** included in the wearable device **101** is not limited to the embodiment. The wearable device **101** may include one or more cameras, such as one or more cameras **1440** that will be described later with reference to FIGS. **14**A to **14**B and/or FIGS. **15**A to **15**B. The wearable device **101** including a plurality of cameras may obtain information on external environment including the wearable device **101** and a user wearing the wearable device **101** from the plurality of cameras arranged in different directions. For example, the wearable device **101** may obtain a second image with respect to the external environment along with a first image with respect to the user's two eyes. The wearable device **101** may identify a portion of the second image that the user gazes in the second image by using the first image.

[0050] According to an embodiment, the sensor **230-1** of the wearable device **101** may generate electronic information that may be processed by the processor **210-1** and/or the memory **215-1** of the wearable device **101** from non-electronic information related to the wearable device **101**. The information may be referred to as sensor data. The sensor **230-1** may include a global positioning system (GPS) sensor for detecting a geographic location of the wearable device **101**, an image sensor, an illumination sensor and/or a time-of-flight (ToF) sensor, and an inertial measurement

unit (IMU) for detecting a physical motion of the wearable device **101**.
[0051] In an embodiment, the communication circuit **240-1** of the wearable device **101** may include a hardware component to support transmission and/or reception of an electrical signal between the wearable device **101** and the external electronic device **120**. For example, the communication circuit **240-1** may include at least one of a MODEM, an antenna, and an optical/electronic (O/E) converter. The communication circuit **240-1** may support transmission and/or reception of the electrical signal by using various types of protocols such as ethernet, local area network (LAN), wide area network (WAN), wireless fidelity (Wi-Fi), Bluetooth, Bluetooth low energy (BLE), ZigBee, long term evolution (LTE), 5G new radio (NR) and/or 6G.
[0052] Although not illustrated, according to an embodiment, the wearable device **101** may include an output means for outputting information in a form other than a visualized form. For example, the wearable device **101** may include a speaker for outputting an acoustic signal. For example, the wearable device **101** may include a motor for providing haptic feedback based on vibration.
[0053] Referring to FIG. **2**, the external electronic device **120** connected or interacting with the wearable device **101** may include at least one of a processor **210-2**, a memory **215-2**, a display **220-2**, a sensor **230-2**, or a communication circuit **240-2**. The processor **210-2**, the memory **215-2**, the display **220-2**, the sensor **230-2**, and the communication circuit **240-2** may be electronically and/or operably coupled with each other by a communication bus **202-2**. The processor **210-2**, the memory **215-2**, the display **220-2**, the sensor **230-2**, and the communication circuit **240-2** of the external electronic device **120** may correspond to the processor **210-1**, the memory **215-1**, the display **220-1**, the sensor **230-1**, and the communication circuit **240-1** of the wearable device **101**. In order to reduce repetition of description, among the descriptions of the processor **210-2**, the memory **215-2**, the display **220-2**, the sensor **230-2**, and the communication circuit **240-2** of the external electronic device **120**, descriptions overlapping the processor **210-1**, the memory **215-1**, the display **220-1**, the sensor **230-1**, and the communication circuit **240-1** of the wearable device **101** may be omitted.
[0054] According to an embodiment, the wearable device **101** may monitor a form of the external electronic device **120** by communicating with the external electronic device **120** having a deformable structure. Referring to FIG. **2**, different exteriors of the external electronic device **120** having a deformable structure are illustrated. The display **220-2** of the external electronic device **120** may include a flexible display for the deformable structure of the external electronic device **120**. For example, the external electronic device **120-1** may include a flexible display with a width longer than a height, and a folding axis F formed along a height direction of the flexible display. For example, the external electronic device **120-2** may include a flexible display with a height longer than a width, and a folding axis F formed along a width direction of the flexible display. The sensor **230-2** of the external electronic device **120** (e.g., the external electronic devices **120-1**, **120-2**) including a straight folding axis F may include a hall sensor. The external electronic device **120** may identify an angle of a housing and/or a flexible display of the external electronic device **120** folded with respect to the folding axis F by using the hall sensor. The external electronic device **120** may transmit information associated with the angle to the wearable device **101** through the communication circuit **240-2**.
[0055] In an embodiment, the wearable device **101** may monitor a form of the external electronic device **120** by communicating with the external electronic device **120** including a flexible display insertable into the housing. Referring to FIG. **2**, different exteriors of the external electronic device with a flexible display insertable into the housing are illustrated, such as external electronic devices **120-3** and **120-4**. For example, the external electronic device **120-3** may include a flexible display with a height longer than a width, and may insert or extract the flexible display into the housing along a direction V parallel to the height. For example, the external electronic device **120-4** may include a flexible display with a width longer than a height, and may insert or extract the flexible display into the housing along a direction H parallel to the width. The sensor **230-2** of the external

electronic device **120** that includes an actuator (e.g., a motor) for inserting a flexible display into a housing may include a sensor to measure rotation of the actuator. Using information obtained by using the sensor **230-2**, the external electronic device **120** may identify information on the form of the flexible display, such as a width, a height, a size, and/or area of the flexible display extracted from the housing. The external electronic device **120** may transmit the information on the form of the flexible display to the wearable device **101** through the communication circuit **240-2**. An operation of the wearable device **101** connected to an external electronic device including a flexible display insertable into a housing, such as the external electronic devices **120-3** and **120-4**, will be described with reference to FIG. **13**.

[0056] Referring to an embodiment of FIG. **2**, in the memory **215-1** of the wearable device **101**, one or more instructions (or commands) indicating calculation and/or operation to be performed by the processor **210-1** of the wearable device **101** on data may be stored. A set of one or more instructions may be referred to as a program, a firmware, an operating system, a process, a routine, a sub-routine, and/or an application. In the following, that an application is installed in an electronic device (e.g., the wearable device **101**) may be that one or more instructions provided in a form of an application are stored in the memory **215-1**, and may mean that the one or more applications are stored in an executable format (e.g., a file with an extension specified by an operating system of the wearable device **101**) by the processor of the electronic device. According to an embodiment, the wearable device **101** may execute one or more instructions stored in the memory **215-1** to perform operations of FIGS. **3**, **6**, **7**A to **7**B, **9**, and/or **10**A to **10**B.

[0057] Referring to FIG. **2**, programs installed in the wearable device **101** may be classified into any one layer of the different layers including an application layer **260**, a framework layer **270**, and/or a hardware abstraction layer (HAL) **250**, based on a target. For example, within the hardware abstraction layer **250**, programs (e.g., driver) designed to target hardware (e.g., the display **220-1**, the camera **225**, the sensor **230-1**, and/or the communication circuit **240-1**) of the wearable device **101** may be classified. For example, within the framework layer **270**, programs (e.g., a gaze tracker **271**, a gesture tracker **272**, a motion tracker **273**, an external space recognizer **274** and/or an external electronic device controller **275**) designed to target at least one of the hardware abstraction layer **250** and/or the application layer **260** may be classified. Programs classified as the framework layer **270** may provide an application programming interface (API) executable by another program.

[0058] Referring to FIG. **2**, within the application layer **260**, a program designed to target a user (e.g., the user **110** of FIG. **1**) who controls the wearable device **101** may be classified. For example, a program classified as the application layer **260** may include at least one of an application **261** for playing and/or streaming video, an application **262** for video conference, an application **263** for browsing media content **280** (e.g., an image and/or video) of the memory **215-1**, or an application **264** for call connection. The embodiment is not limited thereto. For example, the program classified as the application layer **260** may cause execution of a function supported by programs classified as the framework layer **270**, by calling the API.

[0059] Referring to FIG. **2**, the wearable device **101** may process information related to a gaze of a user wearing the wearable device **101** based on execution of the gaze tracker **271** in the framework layer **270**. For example, the wearable device **101** may obtain an image including eye of the user from the camera **225**. The wearable device **101** may identify a direction of the user's gaze by using a position and/or a direction of pupil included in the image.

[0060] Referring to FIG. **2**, the wearable device **101** may identify a motion of a preset body part including hand based on execution of the gesture tracker **272** in the framework layer **270**. For example, the wearable device **101** may obtain an image and/or video including the body part from the camera **225**. The wearable device **101** may identify a gesture performed by the preset body part, by using a motion and/or a posture of the preset body part shown by the image and/or video.

[0061] Referring to FIG. **2**, the wearable device **101** may identify a motion of the wearable device

101 based on execution of the motion tracker **273** in the framework layer **270**. In a state in which the wearable device **101** is worn by the user, the motion of the wearable device **101** may be related to a motion of the user's head. For example, the wearable device **101** may identify a direction of the wearable device **101**, which substantially matches a direction of the head. The wearable device **101** may identify the motion of the wearable device **101** by using sensor data of the sensor **230-1** including the IMU.

[0062] Referring to FIG. **2**, the wearable device **101** may obtain information with respect to an external space that includes the wearable device **101** or is adjacent to the wearable device **101**, based on execution of the external space recognizer **274** in the framework layer **270**. The wearable device **101** may obtain the information by using the camera **225** and/or the sensor **230-1**. Referring to FIG. **2**, in a state in which the external space recognizer **274** is executed, the wearable device **101** may identify a virtual space mapped to the external space by using the information obtained by the external space recognizer **274**. The wearable device **101** may identify a location and/or a direction of the wearable device **101** in the external space, based on the execution of the external space recognizer **274**. For example, the wearable device **101** may perform simultaneous localization and mapping (SLAM) to recognize an external space and a location of the wearable device **101** within the external space, based on execution of the external space recognizer **274** and/or the motion tracker **273**.

[0063] Referring to FIG. **2**, according to an embodiment, the wearable device **101** may execute a function for identifying and/or controlling an external electronic device by using the external electronic device controller **275**. The function may include a function (e.g., pairing) of establishing a communication link between the wearable device **101** and the external electronic device **120** using the communication circuit **240-1**. The function may include a function of monitoring a form of the external electronic device **120** by using the camera **225** and/or the communication link. The function may include a function of moving information outputted through the display **220-1** of the wearable device **101** to the display **220-2** of the external electronic device **120**. The function may include a function of moving information outputted through the display **220-2** of the external electronic device **120** to the display **220-1** of the wearable device **101**. In a state in which the external electronic device controller **275** is executed, the wearable device **101** may execute a function of moving information between the displays **220-1** and **220-2** based on a form of the external electronic device **120**. An operation performed by the wearable device **101** based on execution of the external electronic device controller **275** will be described with reference to FIG. **3**.

[0064] In an embodiment, the information moved or transferred between the wearable device **101** and the external electronic device **120** may include information related with media content **280**, such as music, image, and/or video. The information moved or transferred between the wearable device **101** and the external electronic device **120** may include information for executing at least one of the applications **261**, **262**, **263**, and **264**. The information may include text indicating a location (e.g., a location of network and/or the memory **215-1**) of the media content **280** and/or applications, such as uniform resource identifier (URI). The URI may include a uniform resource location (URL), uniform resource name (URN), Uniform Resource Characteristics (URC), or a combination thereof. In drawings and descriptions described later, the information and/or URI transmitted between the wearable device **101** and the external electronic device **120** are exemplarily described.

[0065] As described above, according to an embodiment, the wearable device **101** may output a user interface (UI) based on a form of the external electronic device **120** on the display **220-1**. The wearable device **101** may control movement (e.g., movement from the wearable device **101** to the external electronic device **120**) of the media content **280** based on changing of form of the external electronic device **120**. Based on the movement, the wearable device **101** may support a user wearing the wearable device **101** to switch a plurality of electronic devices including the external electronic device **120**.

[0066] Hereinafter, according to an embodiment, an operation in which the wearable device **101** identifies change of a form of the external electronic device **120** and performs in response to the change will be described with reference to FIG. **3**.

[0067] FIG. **3** illustrates an example of a flowchart of operations performed by a wearable device, according to an embodiment. The wearable device **101** of FIGS. **1** and **2** may perform an operation of the wearable device described with reference to FIG. **3**. For example, the wearable device **101** and/or the processor **210-1** of FIG. **2** may perform at least one of operations of FIG. **3**. For example, the wearable device may perform at least one of operations of FIG. **3**, based on execution of the external electronic device controller **275** of FIG. **2**.

[0068] Referring to FIG. **3**, according to an embodiment, in operation **310**, the wearable device may obtain information on a form of an external electronic device. The wearable device may obtain an image from a camera (e.g., the camera **225** of FIG. **2**) as an example of the information. The wearable device may identify the external electronic device and a form of the external electronic device from the image. The wearable device may identify the external electronic device and/or the form from the image by performing object recognition with respect to the image. In order to perform the object recognition, the wearable device may include hardware (e.g., a neural processing unit (NPU)) and/or software for driving an artificial neural network. The artificial neural network for performing the object recognition may include a plurality of layers interconnected by an architecture, such as a convolutional neural network (CNN), a recurrent neural network (RNN), and/or a long-short term memory (LSTM).

[0069] In an embodiment, the wearable device may identify an angle of a flexible display folded by a folding axis (e.g., the folding axis F of FIGS. **1** to **2**), based on identifying the flexible display of the external electronic device from the image. In an embodiment, the wearable device may identify a size of the flexible display of the external electronic device extracted from a housing of the external electronic device, from the image.

[0070] In an embodiment, the wearable device may obtain information on a form of the external electronic device from the external electronic device through a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**). The information may be transmitted from the wearable device to the external electronic device through a communication link. The communication link may be established based on a wireless communication protocol such as Bluetooth, Wi-Fi direct, and/or near-field communication (NFC). The information obtained by the wearable device from the external electronic device through the communication circuit may be related to a folded angle of the external electronic device. For example, the information may include a numeric value indicating the angle, or an identifier assigned to the form of the external electronic device distinguished by the angle. In an embodiment, the form (or a state) of the external electronic device may be classified as shown in Table 1 by the angle.

TABLE-US-00001 TABLE 1 Angle range Name of form 0° or more and folded state, closed state, fully folded state 15° or less 15° or more and sub folded state, sub closed state, tent state, acute 80° or less angle state, flipped state, partially closed state 80° or more and right angle state, flex state, half opened state 110° or less 110° or more and sub unfolded state, sub opened state, obtuse angle 160° or less state, partially unfolded state, partially opened state 160° or more and unfolded state, sub opened state, straight angle state, 180° or less fully opened state

[0071] In an embodiment, an intermediate form may include remaining forms distinguished from the folded form and the unfolded form of Table 1. In an embodiment, when the wearable device is connected to an external electronic device including a flexible display insertable into a housing, the information obtained by the wearable device from the external electronic device through a communication circuit may include a size in which the flexible display included in the external electronic device is extracted from the housing of the external electronic device. For example, the information may include an identifier assigned to a form of an external electronic device classified

by the size.
[0072] Referring to FIG. **3**, according to an embodiment, in operation **320**, the wearable device may identify whether the external electronic device has been transformed into a first form. The wearable device may identify whether the external electronic device has been transformed into the first form, by using the information of operation **310**. The first form of operation **320** may be set to move information visualized by the wearable device from the wearable device to the external electronic device. In a state in which an external electronic device including a folding axis is identified, the first form may include a form in which the folding axis is unfolded by exceeding a preset angle (e.g., a preset angle of 180° or less). In a state in which an external electronic device including a flexible display insertable into a housing is identified, the first form may include a form in which the flexible display is extracted from the external electronic device by exceeding a preset size.
[0073] Referring to FIG. **3**, according to an embodiment, in response to identifying that the external electronic device is transformed into the first form by using the information in operation **310** (**320**-YES), in operation **330**, the wearable device may transmit a first signal for displaying at least one first screen using the external electronic device to the external electronic device. The at least one first screen, which is a screen being displayed through a display of the wearable device, may include media content (e.g., the media content **280** of FIG. **2**) or a window provided by an application of the wearable device. For example, the wearable device may transmit the first signal for displaying at least one media content using a flexible display to the external electronic device, based on identifying unfolding of the flexible display of the external electronic device using the information in operation **310**. In the above example, the first signal may include the at least one media content, or may include URL of the at least one media content. While the external electronic device is transformed into the first form, the wearable device may display a visual object for guiding or informing of movement of the at least one first screen.
[0074] According to an embodiment, in a state identifying that the external electronic device is not transformed into the first form (**320**-NO), the wearable device may identify whether the external electronic device has been transformed into a second form, based on operation **340**. The second form of operation **340** may be set to move information visualized by the external electronic device from the external electronic device to the wearable device. In a state in which an external electronic device including folding axis is identified, the second form may include a form in which the folding axis is folded below a preset angle (e.g., a preset angle of 0° or more). In a state in which an external electronic device including a flexible display insertable into a housing is identified, the second form may include a form in which the flexible display is inserted into the external electronic device below a preset size.
[0075] Referring to FIG. **3**, when the external electronic device is not transformed into either the first form or the second form (**340**-NO), the wearable device may maintain monitoring of a form of the external electronic device based on operation **310**.
[0076] Referring to FIG. **3**, according to an embodiment, in response to identifying that the external electronic device is transformed into the second form based on the information in operation **310** (**340**-YES), in operation **350**, the wearable device may transmit a second signal for displaying at least one second screen provided by the external electronic device through the display of the wearable device to the external electronic device. The second signal may include a command and/or a request for identifying media content outputted through the external electronic device and/or at least one application executed by the external electronic device. After transmitting the second signal, in response to receiving information on the at least one second screen from the external electronic device, the wearable device may display the at least one second screen through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device. For example, in response to receiving one or more packets including at least a portion of the media content, or receiving URL of the media content, the wearable device may display the media content. For example, based on

identifying an identifier (e.g., package name) of at least one application executed by the external electronic device, the wearable device may execute the at least one application or display UI for installing the at least one application. While the external electronic device is transformed into the second form, the wearable device may display a visual object for guiding or informing of movement of the at least one second screen.

[0077] As described above, according to an embodiment, the wearable device may control movement of a screen between the wearable device and the external electronic device according to a form of the external electronic device. For example, the wearable device identifying the external electronic device transformed into the first form of operation **320** may move at least one first screen of the display of the wearable device to the external electronic device. For example, the wearable device identifying the external electronic device transformed into the second form of operation **340** may move at least one second screen of the display (e.g., a flexible display) of the external electronic device to the wearable device. The wearable device may recognize a motion of transforming the external electronic device as a gesture for moving a screen between the wearable device and the external electronic device.

[0078] Hereinafter, an operation of the wearable device for the first form of operation **320** and the second form of operation **340** will be described with reference to different drawings. For example, an operation of the wearable device with respect to the external electronic device transformed into the first form of operation **320** will be described with reference to FIGS. **4**, **8**A, **11**A to **11**C, **12** and/or **13**. For example, an operation of the wearable device with respect to the external electronic device transformed into the second form of operation **340** will be described with reference to FIGS. **5**, **8**B, and/or **13**.

[0079] FIG. **4** illustrates an example of an operation in which a wearable device **101** moves media content to an external electronic device **120**, according to an embodiment. The wearable device **101** and the external electronic device **120** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIG. **4**. An operation of the wearable device **101** described with reference to FIG. **4** may be related to at least one of the operations of FIG. **3** (e.g., the operation **320** of FIG. **3**).

[0080] According to an embodiment, referring to FIG. **4**, different states **401**, **402**, and **403** in which the wearable device **101** displays UI through a display (e.g., the display **220-1** of FIG. **2**) while the wearable device **101** is connected to an external electronic device **120** foldable by a straight folding axis F are illustrated. The wearable device **101** may identify the external electronic device **120** by using a camera (e.g., the camera **225** of FIG. **2**) and/or a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**). For example, the wearable device **101** may identify the external electronic device **120** at least partially included in field-of-view (FoV) of a user **110**, by using a camera disposed along a direction of eyes of the user **110** wearing the wearable device **101**.

[0081] Referring to states **401**, **402**, and **403** of FIG. **4**, while the wearable device **101** provides a user experience based on VST, AR, and/or MR, the user **110** may see the external electronic device **120** through a display of the wearable device **101**. For example, when the external electronic device **120** is disposed in front of the user **110** (e.g., a direction in which the user **110**′s eyes face), the wearable device **101** may display an image of a camera disposed facing toward the front to the user **110**. Since the image includes the external electronic device **120**, the user **110** may see the external electronic device **120** through the display of the wearable device **101**. For example, when the wearable device **101** includes a display capable of passing ambient light (e.g., ambient light emitted and/or reflected toward the user **110**'s eyes), the user **110** may see the external electronic device **120** through transparent portion of the display.

[0082] Referring to FIG. **4**, in a state **401** in which the external electronic device **120** viewed through the display using a camera is identified, the wearable device **101** may obtain information on a form of the external electronic device **120**. In the state **401** of FIG. **4**, it is assumed that the wearable device **101** displays a screen **140** for reproducing media content (e.g., the media content

280 of FIG. **2**). The wearable device **101** may display a visual object for controlling reproduction of the media content, such as a play button **141**, on the screen **140**. In the state **401** of FIG. **4**, it is assumed that the external electronic device **120** has a form (e.g., the folded form) folded along a folding axis F. The wearable device **101** may obtain the information by using a signal transmitted from the external electronic device **120** and/or the external electronic device **120** identified by an image of the camera.

[0083] In the state **401** of FIG. **4**, as the user **110** unfolds the external electronic device **120**, the wearable device **101** may switch from state **401** to state **402**. In the state **402**, the wearable device **101** may identify that an angle A**1** of the folding axis F increases by exceeding a first threshold angle (e.g., 15°). In the state **402** in which deformation of the display **220-2** (e.g., flexible display) of the external electronic device **120** is identified by using information on a form of the external electronic device **120**, the wearable device **101** may display a visual object **410** for guiding or informing of movement of the screen **140**. For example, the wearable device **101** may display the visual object **410**, based on identifying the angle A**1** that increases by exceeding the first threshold angle. The visual object **410** may include a line (e.g., a dashed line) having a size and/or location of outline (or boundary line) of the screen **140**. In an embodiment, the visual object **410** may be referred to as a visual cue.

[0084] According to an embodiment, in the state **402** in which the angle A**1** of the folding axis F is increased more than or equal to the first threshold angle, the wearable device **101** may visualize movement of the screen **140** to the external electronic device **120** by using the visual object **410**. For example, as the external electronic device **120** is unfolded, the wearable device **101** may identify that the angle A**1** increases. As the angle A**1** increases, the wearable device **101** may move the visual object **410** from the screen **140** toward the external electronic device **120**. For example, when the angle A**1** of the folding axis F matches the first threshold angle, the wearable device **101** may begin to display the visual object **410** having a size and/or a location matching the outline of the screen **140**. Then, as the angle A**1** continues to increase, the wearable device **101** may move the visual object **410** along a direction D**1**. As the angle A**1** increases, the wearable device **101** may change a size of the visual object **410** according to a size of the external electronic device **120** viewed through the display. In the exemplary state **402** of FIG. **4**, in which the size of the external electronic device **120** viewed through the display is smaller than a size of the screen **140**, as the angle A**1** increases, the wearable device **101** may gradually reduce the size of the visual object **410**.

[0085] According to an embodiment, a form, a size, and/or a location of the visual object **410** displayed by the wearable device **101** are not limited to the exemplary operation of FIG. **4**. The visual object **410** may flicker by a preset interval by the wearable device **101**. The wearable device **101** may guide movement of the screen **140** to the external electronic device **120** by not only displaying the visual object **410**, but also changing the size and/or location of the screen **140**. The wearable device **101** may visualize movement of the screen **140** from the wearable device **101** to the display **220-2** of the external electronic device **120** by using a linear change of the visual object **410** and/or the screen **140** with respect to the angle A**1**. As in the state **402** of FIG. **4**, the wearable device **101** may linearly move the visual object **410** between the screen **140** and the display **220-2** according to the angle A**1**. For example, as the angle A**1** increases from the first threshold angle, the wearable device **101** may gradually move the visual object **410** to the display **220-2**. A mode of the wearable device **101** in which the visual object **410** is displayed, such as the state **402**, may be referred to as a transition mode.

[0086] According to an embodiment, the wearable device **101** may enter the transition mode by using a direction of a gaze of the user **110** wearing the wearable device **101**. The wearable device **101** may identify the direction of the gaze by using an image that is identified through a camera and includes eyes of the user **110**. In a state in which a direction of the gaze toward the screen **140** is identified, the wearable device **101** may display the visual object **410** and enter the transition mode, based on identifying that the angle A**1** of the external electronic device **120** increases above the first threshold angle. For example, in a state in which a direction of the gaze toward a point spaced apart from the screen **140** is identified, the wearable device **101** may restrict displaying the visual object **410**, and/or may not enter the transition mode, independently of the angle A**1** of the external electronic device **120**.

[0087] According to an embodiment, in the state **402** guiding movement of media content reproduced through the screen **140** and/or the screen **140** using the visual object **410**, the wearable device **101** may transmit information for moving the screen **140** and/or the media content to the external electronic device **120**. The information may include media content reproduced through the screen **140**, or may include an address (e.g., URL) in a network of the media content. The wearable device **101** may transmit information (e.g., information on timing of the media content reproduced through the screen **140**) for continuous reproduction of the media content to the external electronic device **120**, together with the address. The wearable device **101** may transmit information on an application related to the screen **140** to the external electronic device **120**. The information may include information for identifying or installing the application. The operation that the wearable device **101** transmits the information to the external electronic device **120** may be performed, in response to identifying an angle A**1** of folding axis F matching a preset threshold angle. The preset threshold angle may be an angle (e.g., 160°) corresponding to the first threshold angle or exceeding the first threshold angle.

[0088] In the state **402** of FIG. **4**, as the user **110** continues to unfold the external electronic device **120**, the wearable device **101** may switch from the state **402** to the state **403**. In the state **403**, the wearable device **101** may display the screen **420** corresponding to the screen **140** through the display **220-2** of the external electronic device **120**, by controlling the external electronic device **120**. The wearable device **101** may enter the state **403** based on whether an angle A**2** of folding axis F corresponds to a preset form (e.g., the first form of FIG. **3**). For example, based on identifying that the angle A**2** is included in a preset angle range including 180° (e.g., straight angle), the wearable device **101** may enter the state **403** and cause the external electronic device **120** to display the screen **420**. The preset angle range may be classified by a threshold angle of 180° or 180° or less. The wearable device **101** entering the state **403** may at least temporarily cease to display the screen **140** via a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101**.

[0089] Referring to the state **403** of FIG. **4**, since display of the screen **140** through the display of the wearable device **101** is stopped and the screen **420** corresponding to the screen **140** is displayed through the display **220-2** of the external electronic device **120**, the user **110** may recognize movement of media content (e.g., media content reproduced through the screen **140**) from the wearable device **101** to the external electronic device **120**. In the state **403** in which movement of the screen **140** from the wearable device **101** to the external electronic device **120** is completed, the wearable device **101** may at least temporarily cease to display the visual object **410** for guiding or informing of movement of the screen **140**.

[0090] In the state **403** in which media content corresponding to the screen **140** is moved from the wearable device **101** to the external electronic device **120**, the wearable device **101** may display the display **220-2** of the external electronic device **120**. For example, a wearable device **101** displaying an image obtained from a camera based on VST may display a portion corresponding to the display **220-2** in the image to a user **110** wearing the wearable device **101**. In order to display the portion to the user **110**, the wearable device **101** may restrict and/or stop synthesizing a virtual object to the portion in the image. For example, a wearable device **101** blocking ambient light based on AR and/or MR may make transparency of a portion between eye of the user **110** wearing the wearable device **101** and the display **220-2** into preset transparency (e.g., substantially 100% transparency), in lens of the wearable device **101**. According to the preset transparency, light emitted from the display **220-2** may pass through the lens of the wearable device **101** and propagate to the user **110**'s eyes. In an embodiment, the wearable device **101** that moves media content corresponding to the screen **140** to the external electronic device **120** may enter a power saving mode (or a sleep mode), and display a visual object (e.g., a pop-up window including preset text such as “Take off HMD”) to guide attachment and detachment of the wearable device **101**.

[0091] As described above, according to an embodiment, the wearable device **101** may identify deformation of the display **220-2** (e.g., flexible display) of the external electronic device **120** by using information on a form of the external electronic device **120**. Based on identifying the deformation, the wearable device **101** may display the visual object **410** for guiding or informing of movement of at least one media content on a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101**. While the visual object **410** is displayed (e.g., the state **402**), the wearable device **101** may execute a function for moving at least one media content between the wearable device **101** and the external electronic device **120**, based on the deformation of the display **220-2**. The function may be executed based on identifying the user **110**'s gaze facing the screen **140** displayed by the wearable device **101**. The user **110** may move the screen **140** displayed through the wearable device **101** to the display **220-2** of the external electronic device **120** distinguished from the wearable device **101**, by executing the function. According to an embodiment, based on movement of the screen **140**, the wearable device **101** may sequentially switch between states **401**, **402**, and **403** based on unfolding of the external electronic device **120**.

[0092] Hereinafter, according to an embodiment, an operation performed by the wearable device **101** based on folding of the external electronic device **120** will be described with reference to FIG. **5**.

[0093] FIG. **5** illustrates an example of an operation in which a wearable device **101** moves media content displayed through an external electronic device **120** to the wearable device **101**, according to an embodiment. The wearable device **101** and the external electronic device **120** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIG. **5**. An operation of the wearable device **101** described with reference to FIG. **5** may be related to at least one of the operations (e.g., the operation **340** of FIG. **3**) of FIG. **3**.

[0094] Referring to FIG. **5**, according to an embodiment, exemplary states **501**, **502**, and **503** in which the wearable device **101** is connected to the external electronic device **120** foldable by a straight folding axis F are illustrated. As shown in the state **403** of FIG. **4**, the state **501** of FIG. **5** may include a state that allows the wearable device **101** to display the display **220-2** of the external electronic device **120**. In the state **501** of FIG. **5**, the wearable device **101** may identify the external electronic device **120** and/or the display **220-2** of the external electronic device **120** viewed to the user **110** through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101**. In the state **501** of FIG. **5**, it is assumed that the external electronic device **120** displays a screen **505** including media content by using the display **220-2**.

[0095] In the state **501** of FIG. **5**, the wearable device **101** may obtain information on a form of the external electronic device **120**. The wearable device **101** may obtain the information by using an image including the external electronic device **120** and/or a communication link between the external electronic device **120** and the wearable device **101**. The wearable device **101** may monitor the form of the external electronic device **120**, using the information. In the state **501** of FIG. **5**, it is assumed that the wearable device **101** identifies an external electronic device **120** in which an angle A**2** of folding axis F is 180°.

[0096] In the state **501** of FIG. **5**, as the external electronic device **120** is folded, the wearable device **101** may switch from the state **501** to the state **502**. In the state **502**, the wearable device **101** may identify that an angle A**3** of folding axis F is reduced to less than a second threshold angle (e.g., 160°). In the state **502** identifying folding axis F folded below the second threshold angle, the wearable device **101** may display a visual object **510** for guiding or informing of movement of the screen **505**. In an embodiment, the visual object **510** may be referred to as a visual cue. Referring to FIG. **5**, a visual object **510** with a form of a quadrangular dashed line is illustrated as an example, but the embodiment is not limited thereto. The visual object **510** may flicker based on a preset interval by the wearable device **101**. In the state **502** displaying the visual object **510**, the wearable device **101** may change a location, form, color, and/or size of the visual object **510**, by using information (e.g., the angle A**3** of folding axis F) on the form of the external electronic device **120**. For example, as the angle A**3** decreases, the wearable device **101** may gradually increase a size of the visual object **510**. For example, as the angle A**3** is reduced, the wearable device **101** may linearly move the visual object **510** toward a preset position (e.g., center) of the display of the wearable device **101**. As shown in the state **502**, a mode of the wearable device **101** in which the visual object **510** is displayed may be referred to as a transition mode.

[0097] According to an embodiment, the wearable device **101** may enter the transition mode according to a direction of the user **110**'s gaze. For example, the wearable device **101** may enter the transition mode based on identifying a direction of a gaze spaced apart from the screen **505** and/or the display **220-2**. For example, the wearable device **101** may enter the transition mode based on identifying a direction of a gaze toward the visual object **510**. For example, when the direction of the gaze toward the screen **505** and/or the display **220-2** is identified, the wearable device **101** may stop and/or restrict displaying of the visual object **510**, and/or may not enter the transition mode.

[0098] In the state **502** guiding movement of media content reproduced through the screen **505** and/or the screen **505** by using the visual object **510**, according to an embodiment, the wearable device **101** may request information related to the screen **505** from the external electronic device **120**. For example, the wearable device **101** may request information for identifying media content reproduced through the screen **505** to the external electronic device **120**. For example, the wearable device **101** may request information to identify at least one application executed by the external electronic device **120** to provide the screen **505**, to the external electronic device **120**. The wearable device **101** may obtain the information by communicating with the external electronic device **120** in the transition mode. In an embodiment, the wearable device **101** may display the visual object **510** or enter the transition mode, based on obtaining information on the screen **505** from the external electronic device **120**.

[0099] In the state **502** of FIG. **5**, as the user **110** continues to fold the external electronic device **120**, the wearable device **101** may switch from the state **502** to the state **503**. In the state **503**, the wearable device **101** may display a screen **520** corresponding to the screen **505** displayed through the display **220-2** through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101**, by controlling the external electronic device **120**. The wearable device **101** may enter the state **503** based on whether the angle A**3** of folding axis F corresponds to a preset form (e.g., the second form of FIG. **3**). For example, the wearable device **101** may display the screen **520** by entering the state **503**, based on identifying that the angle A**3** is included in a preset angle range including 0°. The preset angle range may be classified by 0° and/or a threshold angle of 0° or more. The wearable device **101** entering the state **503** may transmit a signal for stopping at least temporarily display of the screen **505** through the display **220-2** of the external electronic device **120**, to the external electronic device **120**.

[0100] In the state **503** of FIG. **5**, the wearable device **101** may display a screen **520** floating in field of view (FoV) of the user **110**. Since display of the screen **505** through the display **220-2** of the external electronic device **120** is stopped, and the screen **520** corresponding to the screen **505** is displayed through a display of the wearable device **101**, the user **110** may recognize movement of media content (e.g., media content reproduced through the screen **505**) from the external electronic device **120** to the wearable device **101**. In the state **503** in which movement of the screen **505** from the external electronic device **120** to the wearable device **101** is completed, the wearable device **101** may cease to display the visual object **510** for guiding or informing of movement of the screen **404**.

[0101] As described above, according to an embodiment, in the state in which the external electronic device **120** including folding axis F is identified, the wearable device **101** may identify an angle at which a flexible display (e.g., the display **220-2**) of the external electronic device **120** is folded by folding axis F. As described above with reference to FIG. **4**, the wearable device **101** identifying that the angle increases may move information (e.g., the screen **140** and/or media content reproduced through the screen **140** of FIG. **4**) from the wearable device **101** to the external electronic device **120**. As described above with reference to FIG. **5**, the wearable device **101** identifying that the angle is reduced may move information (e.g., the screen **505** and/or media content reproduced through the screen **505** of FIG. **5**) from the external electronic device **120** to the wearable device **101**.

[0102] Hereinafter, according to an embodiment, an operation performed by the wearable device **101** by a direction of the user **110**'s gaze and/or a form of the external electronic device **120** is described with reference to FIG. **6**.

[0103] FIG. **6** illustrates an example of a flowchart of operations performed by a wearable device, according to an embodiment. The wearable device **101** of FIGS. **1** to **2** may include the wearable device of FIG. **6**. An operation of the wearable device described with reference to FIG. **6** may be performed by the wearable device **101** and/or the processor **210-1** of FIG. **2**. The operation of the wearable device of FIG. **6** may be related to at least one of the operations (e.g., the operation **310** of FIG. **3**) of FIG. **3**. The operation of the wearable device of FIG. **6** may be related to the operation of the wearable device **101** of FIGS. **4** to **5**.

[0104] Referring to FIG. **6**, according to an embodiment, in operation **610**, the wearable device may display at least one screen. The at least one screen may be displayed by execution of at least one of applications **261**, **262**, **263**, and **264** classified as the application layer **260** of FIG. **2**. The at least one screen displayed by the wearable device may include media content (e.g., media content **280** of FIG. **2**) stored in a memory (e.g., the memory **215-1** of FIG. **2**) of the wearable device or streamed through a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**).

[0105] Referring to FIG. **6**, according to an embodiment, in operation **620**, the wearable device may identify a gaze toward an external electronic device (e.g., the external electronic device **120** of FIGS. **1** to **5**) viewed through a display (e.g., the display **220-1** of FIG. **2**). The wearable device may identify a gaze of operation **620** by using an image for eyes of a user wearing the wearable device. The wearable device may obtain the image by controlling a camera (e.g., the camera **225** of FIG. **2**). The wearable device may identify the gaze by using a positional relationship between pupil and iris identified from the image.

[0106] Referring to FIG. **6**, according to an embodiment, in operation **630**, the wearable device may determine whether a communication link between the external electronic device and the wearable device is established. In a state in which a direction of a gaze toward the external electronic device is identified based on operation **620**, the wearable device may identify whether the communication link is established. When the communication link of operation **630** is not established (**630**-NO), the wearable device may display a visual object for establishing a communication link based on operation **650**. The visual object may include a pop-up window, text, and/or icon for guiding or informing of establishment of the communication link. In response to an input for selecting the visual object, the wearable device may initiate establishment of the communication link. For example, the wearable device may establish the communication link by using pairing. In an embodiment, the wearable device may perform operation **640** based on establishment of the communication link.

[0107] According to an embodiment, in a state in which the communication link of operation **630** is established (**630**-YES), the wearable device may perform operation **640** to display at least one screen of operation **610** through an external electronic device, by using a gaze and/or a form of the external electronic device. For example, as described above with reference to FIG. **4**, the wearable device may perform operation **640** according to a direction of the gaze toward the at least one screen and deformation of the external electronic device based on folding.

[0108] As described above, according to an embodiment, by using a direction of the gaze of a user wearing the wearable device, the wearable device may identify the user's intention for interacting with an external electronic device. Based on the intention, the wearable device may activate a communication link between the external electronic device and the wearable device. By using the activated communication link, the wearable device may perform movement of information between the external electronic device and the wearable device. Hereinafter, an operation in which the wearable device performs movement of the information based on operation **640** will be described with reference to FIGS. **7**A to **7**B.

[0109] FIGS. **7**A to **7**B illustrate an example of a signal flowchart between a wearable device **101** and an external electronic device **120**, according to an embodiment. The wearable device **101** and the external electronic device **120** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIGS. **7**A to **7**B. At least one of operations of the wearable device **101** of FIGS. **7**A to **7**B may be performed by the wearable device **101** and/or the processor **210-1** of FIG. **2**. At least one of operations of the external electronic device **120** of FIGS. **7**A to **7**B may be performed by the external electronic device **120** and/or the processor **210-2** of FIG. **2**. An operation of the wearable device **101** and the external electronic device **120** of FIG. **7**A may be related to the operation of the wearable device **101** and the external electronic device **120** described with reference to FIG. **4**. An operation of the wearable device **101** and the external electronic device **120** of FIG. **7**B may be related to the operation of the wearable device **101** and the external electronic device **120** described with reference to FIG. **5**.

[0110] Referring to FIGS. **7**A to **7**B, according to an embodiment, in operation **710**, the wearable device **101** may establish a communication link with the external electronic device **120**. The operation **710** may include at least one of the operations of FIG. **6**. The wearable device **101** may establish a communication link of the operation **710** by using a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**). At least one of the operations of FIGS. **7**A to **7**B may be performed in a state in which a communication link of operation **710** is established.

[0111] Referring to FIG. **7**A, according to an embodiment, in operation **712**, the wearable device **101** may identify that an external electronic device **120** in a folded state is unfolded at a preset angle (e.g., a first threshold angle such as 15°). In an embodiment, the wearable device **101** may identify unfolding of the operation **712** based on a form of the external electronic device **120** identified through a camera (e.g., the camera **225** of FIG. **2**). In an embodiment, the wearable device **101** may perform operation **712** by using a communication link of operation **710**. For example, in operation **714**, the external electronic device **120** may transmit information indicating an angle of folding axis to the wearable device **101** through a communication link, based on identifying that the angle of the folding axis of the external electronic device matches the preset angle. Operations **712** and **714** of FIG. **7**A may be related to the operation **310** of FIG. **3**.

[0112] Referring to FIG. **7**A, according to an embodiment, in operation **720**, the wearable device **101** may display a visual object for guiding or informing of movement of at least one screen. In a state in which a direction of a gaze toward at least one screen is identified, the wearable device **101** may perform operation **720**. The visual object may include the visual object **410** of FIG. **4**. According to an embodiment, after displaying the visual object in operation **720**, in operation **722**, the wearable device **101** may identify that the external electronic device **120** is unfolded by exceeding a preset angle. The wearable device **101** may identify unfolding of the external electronic device **120** by using an image including the external electronic device **120**. The embodiment is not limited thereto, and in operation **724**, the external electronic device **120** may identify unfolding of the external electronic device **120**. In response to identification of unfolding, the external electronic device **120** may transmit information indicating unfolding of the external electronic device **120** to the wearable device **101**. The information may be transmitted from the external electronic device **120** to the wearable device **101** based on occurrence of a preset event corresponding to the unfolding. The information may include data on an angle of folding axis measured by the external electronic device **120** and/or a form of the flexible display of the external electronic device **120** classified by the angle. In operations **714** and **724**, the external electronic device **120** may identify an angle of folding axis of the external electronic device **120**, by using data of a sensor (e.g., the sensor **230-2** of FIG. **2**). The external electronic device **120** may perform at least one of the operations **714** and **724**, based on the angle.

[0113] According to an embodiment, in the state of identifying the external electronic device **120** unfolded by exceeding a preset angle, the wearable device **101** may change a visual object by performing operation **730**, according to a form of the external electronic device. For example, the wearable device **101** may change a form, size, and/or location of a visual object of operation **720** by using a form of the external electronic device **120**. As described above with reference to FIG. **4**, as an angle of folding axis of the external electronic device **120** increases, the wearable device **101** may move the visual object in a direction from at least one screen displayed by the wearable device **101** toward the flexible display of the external electronic device **120**. The wearable device **101** may change a size of the visual object according to a size of the flexible display of the external electronic device **120**, based on the increase in the angle.

[0114] As the external electronic device **120** is unfolded, a state of the external electronic device **120** may be switched into an unfolded state. Referring to FIG. **7**A, according to an embodiment, in operation **732**, the wearable device **101** may transmit a signal for displaying at least one screen by using the external electronic device, based on whether the external electronic device **120** is in the unfolded state. The signal may include information for continuously moving the at least one screen being displayed by the wearable device **101** to the external electronic device **120**. For example, the signal may include URL of media content included in the at least one screen displayed by the wearable device **101**. The URL may include a parameter indicating a time interval of the media content reproduced in the wearable device **101**, together with addresses for accessing the media content on a network (e.g., the Internet). For example, the signal may include at least a portion of the media content. For example, the signal may include information for executing and/or installing an application that provides the at least one screen displayed by the wearable device **101**. The wearable device **101** transmitting a signal of operation **732** may at least temporarily cease to display at least one screen corresponding to the signal and/or a visual object of operation **720**.

[0115] Referring to FIG. **7**A, in operation **734**, the external electronic device **120** may display at least one screen provided from the wearable device **101**. In response to receiving a signal of operation **732**, the external electronic device **120** may continuously display the at least one screen displayed by the wearable device **101** as shown in the screen **420** of FIG. **4**. The external electronic device **120** may identify whether the at least one screen displayed through the wearable device **101** includes media content displayable by the external electronic device **120**, by using the signal of operation **732**. The external electronic device **120** may identify at least one application corresponding to the at least one screen displayed through the wearable device **101** by using the signal of operation **732**. When the at least one application is not installed in the external electronic device **120**, the external electronic device **120** may display UI for installing the at least one application.

[0116] In an embodiment of FIG. **7**A, as the external electronic device **120** is unfolded, the external electronic device **120** may display a UI (e.g., a lock screen) for obtaining biometric information, a personal identification number (PIN) and/or a password. The external electronic device **120** may bypass displaying the UI and display at least one screen of operation **734**, based on a user account logged in the wearable device **101** transmitting a signal of operation **732**. For example, when the user account logged in to the wearable device **101** matches a user account registered in the external electronic device **120**, the external electronic device **120** may bypass displaying of the UI. The embodiment is not limited thereto, and the external electronic device **120** may perform operation **734** based on user authentication based on the UI. Alternatively, the external electronic device **120** may bypass displaying the UI and perform operation **734**, based on the signal of operation **732**.

[0117] Referring to FIG. **7**B, in a state in which the communication link of operation **710** is established, the wearable device **101** may identify a gaze toward a preset portion based on operation **750**. The wearable device **101** may perform operation **750** while an external electronic device in an unfolded state is viewed through a display. For example, the wearable device **101** may identify a gaze toward a portion spaced apart from the external electronic device viewed through the display.

[0118] The wearable device **101** identifying a gaze of operation **750** may identify that the external electronic device **120** in unfolded state is folded at a preset angle based on operation **752**. The wearable device **101** may identify deformation (e.g., the folding of the external electronic device **120** in operation **752**) of the external electronic device **120**, by using an image of a camera. The embodiment is not limited thereto, and the wearable device **101** may identify folding of the external electronic device **120** by using information (e.g., data of the sensor **230-2** of FIG. **2**) received from the external electronic device **120** through a communication link. For example, in operation **754**, the external electronic device **120** may identify that folding axis of the external electronic device **120** is folded at a preset angle. The external electronic device **120** identifying that the folding axis is folded at the preset angle may transmit a signal to inform that the external electronic device **120** is folded at the preset angle to the wearable device **101**. The wearable device **101** may perform operation **752**, by using the signal.

[0119] The wearable device **101** identifying that the external electronic device **120** is folded at the preset angle may display a visual object guiding movement of at least one screen displayed on the external electronic device **120** by performing operation **760**. A visual object of operation **760** may include the visual object **510** of FIG. **5**. According to an embodiment, after displaying the visual object of operation **760**, in operation **762**, the wearable device **101** may obtain information on at least one screen displayed on the external electronic device **120**. The wearable device **101** may request the external electronic device **120** to transmit information on the at least one screen by using the communication link of operation **710**. The external electronic device **120** receiving the request may transmit information on at least one screen (e.g., the screen **505** of FIG. **5**) to the wearable device **101**, by performing operation **764**.

[0120] As the external electronic device **120** is folded, the state of the external electronic device **120** may be switched into a folded state. Referring to FIG. **7**B, according to an embodiment, in operation **770**, the wearable device **101** may display at least one screen by using information obtained from the external electronic device **120**, based on whether the external electronic device **120** is in a folded state. For example, the wearable device **101** obtaining information of operation **762** may identify whether at least one screen displayed by the external electronic device **120** may be displayed by the wearable device **101**, by using the information. For example, the wearable device **101** may identify whether media content included in the at least one screen may be outputted by the wearable device **101**. The wearable device **101** may identify whether the media content may be outputted by the wearable device **101** by using a format of the media content and/or a type of application capable of executing the media content. For example, the wearable device **101** may identify whether at least one application providing at least one screen displayed by the external electronic device **120** is installed in the wearable device **101**, by using information of operation **762**. When the at least one application is not installed in the wearable device **101**, the wearable device **101** may display UI for installing the at least one application. After the at least one application is installed by the UI, the wearable device **101** may display the at least one screen based on operation **770**.

[0121] Referring to FIGS. **7**A to **7**B, at least one screen may be moved between the wearable device **101** and the external electronic device **120** according to deformation of the external electronic device **120**. In order to control the movement, the wearable device **101** may monitor a form of the external electronic device **120** by using a camera of the wearable device **101** and/or a communication link of the operation **710**. In order to control the movement, the wearable device **101** may request information on at least one screen displayed through a display (e.g., the display **220-2** of FIG. **2**) of the external electronic device **120**, by transmitting information on at least one screen displayed through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101** to

the external electronic device **120**, or communicating with the external electronic device **120**.
[0122] As described above with reference to FIGS. **4** to **6**, **7**A to **7**B, the wearable device **101** may control movement of at least one screen according to a direction of a gaze of a user wearing the wearable device **101**, together with a form of the external electronic device **120**. The embodiment is not limited thereto, and the wearable device **101** may control movement of the at least one screen according to a positional relationship between the external electronic device **120** and the wearable device **101**. Hereinafter, according to an embodiment, an operation in which the wearable device **101** moves at least one screen between the external electronic device **120** and the wearable device **101** by using the positional relationship will be described with reference to FIGS. **8**A to **8**B, **9**, **10**A to **10**B.
[0123] FIGS. **8**A to **8**B illustrate an example of an operation performed by a wearable device **101** by using a form of an external electronic device **120**, according to an embodiment. The wearable device **101** and the external electronic device **120** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIGS. **8**A to **8**B. Operation of the wearable device **101** described with reference to FIG. **8**A may be related to at least one of the operations of FIG. **3** (e.g., operation **320** of FIG. **3**). Operation of the wearable device **101** described with reference to FIG. **8**B may be related to at least one of the operations of FIG. **3** (e.g., operation **340** of FIG. **3**).
[0124] Referring to FIG. **8**A, according to an embodiment, different states **801**, **802**, and **803** in which the wearable device **101** displays UI through a display (e.g., the display **220-1** of FIG. **2**) while the wearable device **101** is connected to the external electronic device **120** foldable by a straight folding axis F are illustrated. The wearable device **101** worn on a head of the user **110** may identify the external electronic device **120** by using a camera disposed along a direction of eyes of the user. The wearable device **101** may establish a communication link with the external electronic device **120** based on at least one of the operations of FIG. **6**. The wearable device **101** may obtain information on a form of the external electronic device **120** through the camera and/or the communication link.
[0125] Referring to FIG. **8**A, in a state **801** in which the external electronic device **120** folded by folding axis F is identified, the wearable device **101** may display a screen **140** for reproducing media content (e.g., the media content **280** of FIG. **2**). In the state **801** in which the screen **140** is displayed, the wearable device **101** may identify a position of the external electronic device **120** with respect to the screen **140**. The position of the external electronic device **120** with respect to the screen **140** may be identified by a position of the external electronic device **120** identified through a camera of the wearable device **101**. For example, the wearable device **101** may identify whether the external electronic device **120** overlaps the screen **140** by using a distance between the external electronic device **120** and the screen **140** viewed through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101**. For example, the wearable device **101** may enter a transition mode, based on identifying that a distance between a point (e.g., a center of the screen **140**) of the screen **140** and a point (e.g., a center of the external electronic device **120** viewed through a display) of the external electronic device **120** on the display of the wearable device **101** is less than a preset threshold. As shown in the state **801** of FIG. **8**A, based on identifying that at least a portion of the external electronic device **120** overlaps the screen **140** on the display of the wearable device **101**, the wearable device **101** may execute a function for moving the screen **140** to the external electronic device **120** according to a form of the external electronic device **120**.
[0126] Referring to FIG. **8**A, in the state **801** in which the external electronic device **120** is at least partially overlapped with the screen **140**, the wearable device **101** may enter the state **802** based on identifying unfolding of the external electronic device **120**. In the state **802** of identifying that an angle A**1** of folding axis F increases by exceeding a first threshold angle (e.g., 15°), the wearable device **101** may display a visual object **810** based on a transition mode. Although the visual object **810** including a quadrangular dashed line is exemplary illustrated, the embodiment is not limited thereto. According to an embodiment, the wearable device **101** may change a position, a form,

color, and/or a size of the visual object **810** according to the angle A**1** of folding axis F. Similar to the visual object **410** of FIG. **4**, the wearable device **101** may move the visual object **810** to the external electronic device **120** based on the increase in the angle A**1**. As the angle A**1** increases, the wearable device **101** may adjust a size of the visual object **810** to a size of the display **220-2** of the external electronic device **120**. As the angle A**1** is reduced, the wearable device **101** may adjust a size of the visual object **810** to a size of the screen **140**.

[0127] According to an embodiment, in the state **802** guiding movement of the screen **140** and/or media content included in the screen **140** by using the visual object **810**, the wearable device **101** may transmit information for moving the screen **140** and/or the media content to the external electronic device **120**. For example, in response to identifying that the angle A**1** of folding axis F matches another threshold angle (e.g., 160°) exceeding the first threshold angle, the wearable device **101** may transmit the information to the external electronic device **120**. The wearable device **101** may insert at least a portion of the media content into payload of a packet transmitted to the external electronic device **120**. The wearable device **101** may insert URL of the media content into the packet. The wearable device **101** may insert information (e.g., information on a timing of the media content reproduced by the wearable device **101**) for continuous reproduction of the media content, together with the URL. The wearable device **101** may transmit an identifier of an application corresponding to the screen **140** and/or an identifier (e.g., an activity name) of a window corresponding to the screen **140** from among a plurality of windows provided from the application, to the packet.

[0128] In the state **802** guiding movement of the screen **140** by using the visual object **810**, the wearable device **101** may cease to move media content to the external electronic device **120**, based on reduction of the angle A**1** of folding axis F. In the state **802**, the wearable device **101** may cease to move media content to the external electronic device **120**, based on identifying the external electronic device **120** that is moved to a location different from the screen **140**. When the media content is not moved according to reduction of the angle A**1** of folding axis F and/or movement of the external electronic device **120**, the wearable device **101** may cease to display the visual object **810** by switching to the state **801**.

[0129] In response to identifying that external electronic device **120** is fully unfolded, the wearable device **101** may switch from the state **802** guiding movement of the screen **140** by using the visual object **810** to the state **803**. The wearable device **101** may enter the state **803**, based on identifying that an angle A**3** of folding axis F substantially matches 180°. In the state **803**, the wearable device **101** may display a screen **820** corresponding to the screen **140** displayed through the wearable device **101** before the state **803**, through the display **220-2** of the external electronic device **120**, by controlling the external electronic device **120**. In the state **803** of controlling the external electronic device **120** to display the screen **820**, the wearable device **101** may at least temporarily cease to display the screen **140** and/or the visual object **810**.

[0130] In the state **803** in which media content is moved from the wearable device **101** to the external electronic device **120** by controlling visibilities of the screens **140** and **810**, the wearable device **101** may allow displaying the external electronic device **120** through the display of the wearable device **101**. For example, in an image displayed on the display of the wearable device **101**, the wearable device **101** may stop and/or restrict synthesizing a portion in which the external electronic device **120** is captured with a virtual object. As synthesis of the virtual object is stopped, the wearable device **101** may form a punctured region in VR and display the external electronic device **120** through the region.

[0131] As described above, in a state in which the external electronic device **120** is viewed through a preset portion for displaying media content, such as the screen **140**, the wearable device **101** may identify deformation of the external electronic device **120** by using information on a form of the external electronic device **120**. Based on the deformation, the wearable device **101** may control movement of the media content. The wearable device **101** may transmit information for continuously outputting the media content to the external electronic device **120**. While identifying the deformation of the external electronic device **120**, the wearable device **101** may display a visual object **810** for guiding or informing of movement of the media content. The user **110** wearing the wearable device **101** may recognize media content moved from the wearable device **101** to the external electronic device **120**, based on sequential switching of states **801**, **802**, and **803** of FIG. **8**A. Similar to operation of the wearable device **101** based on unfolding of the external electronic device **120** described with reference to FIG. **8**A, the wearable device **101** may identify folding of the external electronic device **120**.

[0132] Referring to FIG. **8**B, according to an embodiment, while the external electronic device **120** including folding axis F is unfolded, different states **804**, **805**, and **806** of the UI displayed by the wearable device **101** are illustrated. In the state **804**, the wearable device **101** may identify an external electronic device **120** having an angle A**2** of folding axis F substantially matched to 180°. The wearable device **101** may identify the angle A**2** based on object recognition of an image of a camera (e.g., the camera **225** of FIG. **2**), or identify the angle A**2** through a communication link between the wearable device **101** and the external electronic device **120**. The wearable device **101** may identify the screen **820** displayed through the display **220-2** of the external electronic device **120** and/or media content included in the screen **820**.

[0133] Referring to FIG. **8**B, in the state **804** of identifying that the external electronic device **120** is fully unfolded, the wearable device **101** may determine whether to enter a transition mode by using a location of the external electronic device **120**. For example, the wearable device **101** may enter the transition mode, based on whether the external electronic device **120** viewed through a display (e.g., the display **220-2** of FIG. **2**) of the wearable device **101** overlaps a preset portion **830** of the display. An exemplary state **804** in which a preset portion **830** is formed on a left side of a displaying region formed by the display is illustrated, but the embodiment is not limited thereto. The wearable device **101** may adjust or customize a position, size, and/or form of the portion **830** based on a user input. In an embodiment, the wearable device **101** may display that the external electronic device **120** overlaps the portion **830** by adjusting transparency and/or color of the portion **830**.

[0134] In the state **804** of FIG. **8**B, in response to identifying the external electronic device **120** overlapped to the portion **830** of the display of the wearable device **101**, the wearable device **101** may enter the state **805**. For example, the wearable device **101** may switch to the state **805** based on identifying the external electronic device **120** overlapped by exceeding the portion **830** and a preset time. While the external electronic device **120** overlaps the portion **830**, the wearable device **101** may execute a function for moving the screen **820** of the external electronic device **120** to the wearable device **101**, based on a form of the external electronic device **120**. Based on entering the state **805**, the wearable device **101** may display a visual object **840** for guiding or informing of movement of the screen **820**. Similar to the visual object **510** of FIG. **5**, the wearable device **101** may adjust a size, a form, color, and/or a position of the visual object **840** by using the angle A**3** of folding axis F of the external electronic device **120**. For example, as the angle A**3** is reduced, the wearable device **101** may move the visual object **840** in a direction away from the external electronic device **120** viewed through the display of the wearable device **101**. For example, as the angle A**3** is reduced, the wearable device **101** may gradually change a size of the visual object **840** to a size (e.g., a size of the screen **850**) different from the size of the display **220-2** and/or the screen **820**. In the state **805**, the wearable device **101** may obtain and/or request information for moving the screen **820** from the external electronic device **120**.

[0135] In the state **805** of guiding movement of the screen **820** by using the visual object **840**, the wearable device **101** may monitor deformation of the external electronic device **120**. The wearable device **101** may switch from the state **805** to the state **806**, based on identifying that the angle A**3** of folding axis F of the external electronic device **120** matches substantially 0°. In the state **806**, the external electronic device **120** may be fully folded. In the state **806** identifying that the external electronic device **120** is fully folded, the wearable device **101** may display a screen **850** corresponding to the screen **820** that was displayed on the display **220-2** of the external electronic device **120** through the display of the wearable device **101**. For example, the screen **850** displayed by the wearable device **101** may include media content included in the screen **820** that was displayed by the external electronic device **120**. A first timing of the media content reproduced through the screen **850** may match a second timing of the media content reproduced through the screen **820**, or may be after the second timing. For example, the screen **850** may be provided by an application that was executed by the external electronic device **120** to provide the screen **820**. In the state **806** displaying the screen **850**, the wearable device **101** may transmit a signal for informing display of the screen **850** or switching a mode of the external electronic device **120** to a preset mode (e.g., lock mode and/or sleep mode), to the external electronic device **120**.

[0136] As described above, according to an embodiment, the wearable device **101** may move the screen **820** displayed through the external electronic device **120** to the wearable device **101**, based on folding of the external electronic device **120**. Based on the movement, the wearable device **101** may continuously display information of the display **220-2** that is covered by folding of the external electronic device **120**. Since the wearable device **101** performs movement of the screen **820** in response to deformation of the external electronic device **120**, the wearable device **101** may not require an input different from the deformation.

[0137] Hereinafter, according to an embodiment, an operation performed by the wearable device **101** based on a location and/or a form of the external electronic device **120** will be described with reference to FIG. **9**.

[0138] FIG. **9** illustrates an example of a flowchart of operations performed by a wearable device, according to an embodiment. The wearable device **101** of FIGS. **1** to **2** may include the wearable device of FIG. **9**. An operation of the wearable device described with reference to FIG. **9** may be performed by the wearable device **101** and/or the processor **210-1** of FIG. **2**. An operation of the wearable device of FIG. **9** may be related to at least one (e.g., the operation **310** of FIG. **3**) of the operations of FIG. **3**. An operation of the wearable device of FIG. **7** may be related to the operation of the wearable device **101** of FIG. **9**.

[0139] Referring to FIG. **9**, according to an embodiment, in operation **910**, the wearable device may display at least one screen. The wearable device may perform operation **910** similar to the operation **610** of FIG. **6**. For example, the wearable device may display at least one screen of the operation **910**, based on execution of at least one application installed in the wearable device. The wearable device may display at least one screen including media content stored in a memory (e.g., the memory **215-1** of FIG. **2**) or streamed through a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**).

[0140] Referring to FIG. **9**, according to an embodiment, in operation **920**, the wearable device may identify that the external electronic device is viewed through a preset position of the display (e.g., the display **220-1** of FIG. **2**). The preset position may be included in a portion where at least one screen of operation **910** is viewed in a displaying region formed by the display of the wearable device. The preset position may be included in a region set on the displaying region for movement of information between the wearable device and the external electronic device, as shown in the portion **830** of FIG. **8**B.

[0141] According to an embodiment, in a state of identifying the external electronic device viewed through the preset position of the display of the wearable device, the wearable device may identify whether a communication link between the external electronic device and the wearable device is established, by performing operation **930**. The wearable device may perform operation **930** of FIG. **9** similar to operation **630** of FIG. **6**. When the communication link of operation **930** is not established (**930**-NO), the wearable device may display a visual object for establishing a communication link, based on operation **950**. The visual object may include a pop-up window, text, and/or an icon to guide establishment of the external electronic device and/or the communication link. The wearable device may perform operation **950** of FIG. **9** similar to operation **650** of FIG. **6**. For example, in response to an input related to the visual object, the wearable device may establish a communication link with the external electronic device. In response to establishment of the communication link, the wearable device may perform operation **940**.

[0142] According to an embodiment, in a state in which the communication link of operation **930** is established (**930**-YES), the wearable device may move at least one screen to the external electronic device, based on a position where the external electronic device is viewed through a display, by performing operation **940**. As described above with reference to FIG. **8**A, the wearable device may identify deformation of the external electronic device at least partially overlapped with at least one screen of operation **910**. The wearable device identifying the deformation may transmit information for moving at least one screen of operation **910** to the external electronic device. The information may include data (e.g., identifier of an application providing the at least one screen, and/or URL of at least one media content included in the at least one screen) for continuously displaying the at least one screen by using the external electronic device. Movement of the at least one screen of operation **940** may be performed by using a communication link established based on operations **930** and **950**. For example, the exemplified data may be transmitted from the wearable device to the external electronic device through the communication link.

[0143] As described above, according to an embodiment, by using a motion (e.g., a user's gaze and/or a user's motion with respect to the external electronic device) of a user wearing the wearable device, the wearable device may identify the user's intention for moving information between the wearable device and the external electronic device. For example, in response to identifying a gaze toward a screen to be moved from the wearable device to the external electronic device, and/or the external electronic device overlapped on the screen, the wearable device may move the screen to the external electronic device based on deformation of the external electronic device. Hereinafter, referring to FIGS. **10**A to **10**B, an operation in which the wearable device moves at least one screen based on operation **940** will be described.

[0144] FIGS. **10**A to **10**B illustrate an example of a signal flowchart between a wearable device **101** and an external electronic device **120**, according to an embodiment. The wearable device **101** and the external electronic device **120** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIGS. **10**A to **10**B. At least one of operations of the wearable device **101** of FIGS. **10**A to **10**B may be performed by the wearable device **101** and/or the processor **210-1** of FIG. **2**. At least one of operations of the external electronic device **120** of FIGS. **10**A to **10**B may be performed by the external electronic device **120** and/or the processor **210-2** of FIG. **2**. FIGS. **10**A to **10**B may correspond to FIGS. **8**A to **8**B, respectively.

[0145] Referring to FIGS. **10**A to **10**B, according to an embodiment, in operation **1010**, the wearable device **101** may establish a communication link. The operation **1010** may include at least one of the operations of FIGS. **6** and/or **9**. The wearable device **101** may establish a communication link of the operation **1010**, by communicating with the external electronic device **120** by using a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**). At least one of the operations of FIGS. **10**A to **10**B may be performed in a state in which the communication link of the operation **1010** is established.

[0146] Referring to FIG. **10**A, according to an embodiment, in operation **1012**, the wearable device **101** may identify an external electronic device viewed through a preset region of a display (e.g., the display **220-1** of FIG. **2**). The preset region in operation **1012** may include at least one screen of the wearable device **101** and/or a portion where at least one media content is displayed. For example, in the state **801** of FIG. **8**A, the preset region in operation **1012** may include a portion of a displaying region in which the screen **140** is disposed.

[0147] According to an embodiment, in a state in which the external electronic device **120** viewed through the preset region in operation **1012** is identified, the wearable device may identify that the external electronic device **120** in a folded state is unfolded at a preset angle in the operation **1020**.

For example, in response to identifying the external electronic device **120** that is viewed by exceeding the preset time in a preset region, the wearable device **101** may identify unfolding of the operation **1020**. The wearable device **101** may obtain information on a form of the external electronic device **120** by using a camera (e.g., the camera **225** of FIG. **2**) and/or a communication link of the operation **1010**. The information may include a numeric value indicating an angle of folding axis (e.g., the folding axis F of FIG. **8**A) included in the external electronic device **120**. The wearable device **101** may identify unfolding of the external electronic device **120**, based on identifying that the angle increases by exceeding **0**° and/or a threshold angle exceeding 0°.
[0148] Referring to FIG. **10**A, in operation **1022**, the external electronic device **120** may identify that an angle of folding axis of the external electronic device **120** matches a preset angle. The external electronic device **120** may monitor an angle of folding axis of the external electronic device **120** by using a sensor (e.g., the sensor **230-2** of FIG. **2**). Based on identifying that a monitored angle increases by exceeding 0° and/or a threshold angle exceeding 0°, the external electronic device **120** may transmit a signal indicating unfolding of the external electronic device **120** to the wearable device **101**. The wearable device **101** receiving the signal may identify unfolding of operation **1020**.
[0149] Referring to FIG. **10**A, according to an embodiment, in operation **1030**, the wearable device **101** may display a visual object for guiding or informing of movement of at least one screen. The wearable device **101** may perform operation **1030** based on identifying unfolding of the external electronic device **120** of operation **1020**. A visual object in operation **1030** may include the visual object **810** of FIG. **8**A. According to an embodiment, after displaying the visual object in operation **1030**, the wearable device **101** may identify that the external electronic device **120** is unfolded by exceeding a preset angle, by performing operation **1032**. The preset angle of operation **1030** may be an angle exceeding 0° and include a threshold angle for distinguishing an unfolded state of the external electronic device **120**.
[0150] In an embodiment, unfolding of the external electronic device **120** may be identified by the external electronic device **120** as well as the wearable device **101**. For example, in operation **1034**, the external electronic device **120** may identify that the external electronic device **120** is unfolded, by using a sensor of the external electronic device **120**. The external electronic device **120** identifying unfolding of operation **1034** may transmit a signal indicating unfolding of the external electronic device **120** to the wearable device **101**. The wearable device **101** receiving the signal may identify that the external electronic device **120** is unfolded in operation **1032**. The signal may include an angle of folding axis measured by the external electronic device **120**.
[0151] According to an embodiment, in a state identifying that the external electronic device **120** is unfolded by exceeding of the preset angle, the wearable device **101** may change a visual object of operation **1030** according to a position of the external electronic device **120**, by performing operation **1040**. While the external electronic device **120** is unfolded, the wearable device **101** may guide movement of at least one screen based on the unfolding, by changing a position, a form, color, and/or a size of the visual object of operation **1030**. For example, as the external electronic device **120** is unfolded, the wearable device **101** may reduce a distance between the external electronic device **120** and the visual object. For example, as the external electronic device **120** is folded, the wearable device **101** may increase the distance.
[0152] Referring to FIG. **10**A, according to an embodiment, in operation **1042**, the wearable device **101** may transmit a signal for displaying at least one screen by using an external electronic device, based on whether the external electronic device is in unfolded state. The wearable device **101** may transmit a signal of operation **1042** to the external electronic device **120**, based on identifying that an angle of folding axis of the external electronic device **120** substantially matches 180°. The signal may include information for continuously moving at least one screen displayed by the wearable device **101** to the external electronic device **120**. For example, the signal may include information on at least one media content included in the at least one screen (e.g., a portion of the media content

and/or URL in which the media content is stored). The wearable device **101** may transmit the signal including an identifier of at least one application executed by the wearable device **101** to provide at least one screen. In operation **1042**, the wearable device **101** may at least temporarily cease to display at least one screen. While the external electronic device **120** is unfolded, the at least one screen may be displayed through the external electronic device **120** instead of the wearable device **101**.

[0153] Referring to FIG. **10**A, in operation **1044**, the external electronic device **120** may display at least one screen provided from the wearable device. In response to receiving a signal transmitted from the wearable device **101** in operation **1042**, the external electronic device **120** may display at least one screen corresponding to the signal through a display (e.g., the display **220-2** of FIG. **2**) of the external electronic device **120**. As the external electronic device **120** is unfolded according to operation **1024**, a display (e.g., flexible display) of the external electronic device **120** may be viewed to a user wearing the wearable device **101**. In operation **1044**, the wearable device **101** may display at least one screen being displayed by the wearable device **101** through the display, which is viewed by the user, of the external electronic device **120**. Therefore, the user of the wearable device **101** may view the at least one screen through the display of the external electronic device **120**.

[0154] Referring to FIG. **10**A, in order to display at least one screen in operation **1044**, the external electronic device **120** may perform one or more operations for unlocking the external electronic device **120**. For example, the external electronic device **120** may display UI for inputting biometric information, PIN, and/or password through a display of the external electronic device **120**. In an embodiment, the external electronic device **120** may bypass performing operation of authenticating a user, by display the UI or using the UI, based on a signal of operation **1042**. In an embodiment, the external electronic device **120** may perform operation **1044** more quickly, in response to unfolding of the external electronic device **120**, based on the bypass. In an embodiment, when one user is commonly logged in to the wearable device **101** transmitting the signal and the external electronic device **120**, the external electronic device **120** may bypass display of the UI and perform operation **1044**.

[0155] Referring to FIG. **10**B, according to an embodiment, in a state in which a communication link of operation **1010** is established, the wearable device **101** may identify an external electronic device **120** viewed through a preset region in operation **1050**. The preset region of operation **1050** may include the portion **830** of FIG. **8**B. The wearable device **101** identifying the external electronic device **120** viewed through the preset region may identify that the external electronic device in unfolded state is folded at a preset angle in operation **1052**. The wearable device **101** may identify deformation of the external electronic device **120**, by using an image with respect to the external electronic device **120** at least partially overlapping the preset region. The preset angle of operation **1052** may be included in an angle range (e.g., a preset angle range including 0°) for distinguishing a folded state of the external electronic device **120**.

[0156] Referring to FIG. **10**B, in operation **1054**, the external electronic device **120** may identify that folding axis of the external electronic device **120** is folded at a preset angle. The external electronic device **120** may identify the external electronic device **120** is folded, by using a sensor (e.g., the sensor **230-2** of FIG. **2**, including a hall sensor) for measuring the angle of the folding axis. Based on identifying that the external electronic device **120** is folded, the external electronic device **120** may transmit a signal for informing folding of the external electronic device **120** to the wearable device **101**. The wearable device **101** may perform operation **1052** by using the signal.

[0157] Referring to FIG. **10**B, according to an embodiment, in operation **1060**, the wearable device **101** may display a visual object guiding movement of at least one screen displayed in the external electronic device **120**. The visual object of operation **1060** may include the visual object **840** of FIG. **8**B. While identifying that the external electronic device **120** is folded, the wearable device **101** may guide movement of the at least one screen, by changing the visual object in operation

1060. For example, as an angle of folding axis of the external electronic device **120** is reduced, the wearable device **101** may increase a distance between the visual object and the external electronic device **120**. As the angle of folding axis of the external electronic device **120** increases, the wearable device **101** may reduce the distance. Based on display of the visual object of operation **1060**, the wearable device **101** may enter a transition mode.

[0158] Referring to FIG. **10**B, according to an embodiment, in operation **1062**, the wearable device **101** may obtain information on at least one screen displayed through an external electronic device. The wearable device **101** may request the information on the at least one screen from the external electronic device **120** by using the communication link of operation **1010**. In response to request for the information, the external electronic device **120** may transmit information for displaying at least one screen by perform operation **1064**. The information may include data for continuously displaying at least one screen displayed through a display (e.g., the display **220-2** of FIG. **2**) of the external electronic device **120** by using the wearable device **101**. For example, the information may include data on URL and/or reproduction time of media content included in the at least one screen. For example, the information may include data on at least one application executed by the external electronic device **120** to provide the at least one screen.

[0159] After operation **1052**, the external electronic device **120** may be switched into a folded state by folding axis substantially folded at 0°. In operation **1070** of FIG. **10**B, according to an embodiment, the wearable device **101** may display at least one screen based on whether the external electronic device is in a folded state. The at least one screen of operation **1070** may be related to information obtained by using operation **1062**. For example, the wearable device **101** may output at least one screen displayed through the external electronic device **120**, through a display of the wearable device **101**, by performing operation **1070**. When information obtained by operation **1062** includes data on URL and/or reproduction time of the media content included in the at least one screen of the external electronic device **120**, the wearable device **101** may reproduce the media content by using the information. When information obtained by operation **1062** includes data on at least one application executed by the external electronic device **120** to provide the at least one screen, the wearable device **101** may execute the at least one application or display UI for installing the at least one application.

[0160] As described above with reference to FIGS. **8**A to **8**B, **9**, and **10**A to **10**B, the wearable device **101** may control movement of at least one screen, by using a position of the external electronic device **120** in a displaying region formed by a display of the wearable device **101** together with a form of the external electronic device **120**. The wearable device **101** may move the screen by using a location and/or transformation of the external electronic device **120** without requiring a gesture set for moving the screen between the external electronic device **120** and the wearable device **101**.

[0161] Hereinafter, according to an embodiment, an operation in which the wearable device **101** moves one or more screens to the external electronic device **120** will be described with reference to FIGS. **11**A to **11**C.

[0162] FIGS. **11**A to **11**C illustrate an example of an operation in which a wearable device **101** moves at least one application by using a form of an external electronic device **120**, according to an embodiment. The wearable device **101** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIGS. **11**A to **11**C. An operation of the wearable device **101** described with reference to FIGS. **11**A to **11**C may be related to at least one (e.g., operation **320** of FIG. **3**) of the operations of FIGS. **3**.

[0163] Referring to FIG. **11**A, an exemplary state **1101** in which the wearable device **101** is connected to the external electronic device **120** foldable by a straight folding axis F is illustrated. In the state **1101**, a user **110** wearing the wearable device **101** may view the external electronic device **120** through a displaying region formed through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101**. In the state **1101**, the wearable device **101** may identify the external electronic device **120** in the folded state, through a camera (e.g., the camera **225** of FIG. **2**) and/or a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**).

[0164] In the state **1101**, the wearable device **101** may display one or more screens **1111**, **1112**, and **1113** in FoV of the user **110** wearing the wearable device **101** by executing one or more applications. For example, the wearable device **101** may display the screen **1111** having a widget form based on execution of an application (e.g., a weather application) for visualizing weather information. For example, the wearable device **101** may display the screen **1112** including a region in which video is reproduced based on execution of an application (e.g., the application **261** of FIG. **2**) for streaming video. For example, the wearable device **101** may display the screen **1113** based on execution of an application (e.g., the application **262** of FIG. **2**) for video conference. The state **1101** of FIG. **11**A is exemplary, and the embodiment is not limited thereto.

[0165] According to an embodiment, the wearable device **101** may identify a direction of a gaze of a user **110** wearing the wearable device **101**. The wearable device **101** may obtain information (e.g., an image and/or video in which two eyes are captured) indicating the direction of the gaze by using a camera facing the two eyes of the user **110**. Using the information, the wearable device **101** may estimate a point at which the user **110** gazes within the user **110**'s FoV. As described above with reference to FIGS. **4** to **6**, **7**A to **7**B, the wearable device **101** may control movement of a screen between the wearable device **101** and the external electronic device **120**, based on the direction of the user **110**′s gaze and deformation of the external electronic device **120**.

[0166] Referring to FIG. **11**A, in the state **1101** in which the external electronic device **120** in folded state is identified, the wearable device **101** may identify deformation of the external electronic device **120**. For example, the wearable device **101** may identify that an angle of folding axis F of the external electronic device **120** increases, by using a camera and/or a communication circuit. While the external electronic device **120** is deformed, the wearable device **101** may move a specific screen in which the user **110**'s gaze faces among the screens **1111**, **1112**, and **1113**, to a display (e.g., the display **220-2** of FIG. **2**) of the external electronic device **120**.

[0167] Referring to FIG. **11**A, in the state **1101**, the wearable device **101** that identifies a gaze facing a point P**1** within the screen **1111** may transmit information for displaying the screen **1111** to the external electronic device **120**, based on unfolding of the external electronic device **120**. The state **1102** of the external electronic device **120** of FIG. **11**A may be an exemplary state of the external electronic device **120** receiving information for displaying the screen **1111**. When the external electronic device **120** is folded, the external electronic device **120** may operate in a preset mode (e.g., lock mode) that requires user authentication to activate the external electronic device **120**. After entering the state **1102**, in response to the unfolding, the external electronic device **120** in the lock mode may display a screen for releasing the lock mode. Referring to state **1102**, the external electronic device **120** may display a screen for receiving a PIN and/or a password. The embodiment is not limited thereto, and the external electronic device **120** may display a screen including text and/or icon guiding an input of biometric information.

[0168] Referring to FIG. **11**A, in the state **1102** operating based on the lock mode, the external electronic device **120** may switch to a state **1105** by using information for releasing the lock mode. The information for releasing the lock mode may include at least one of PIN, password, and/or biometric information (e.g., iris, fingerprint, and/or vein pattern). In the state **1105**, the external electronic device **120** may execute an application corresponding to the screen **1111**, by using information received from the wearable device **101**. Referring to FIG. **11**A, in the state **1105** in which information for displaying the screen **1111** corresponding to an application for providing weather information is received, the external electronic device **120** may display a screen provided from the application through a display of the external electronic device **120**. The information received to display the screen **1111** from the wearable device **101** may include an identifier (e.g., package name) of the application executed by the wearable device **101** to provide weather information and/or an identifier of a window corresponding to the screen **1111** among a plurality of windows provided from the application.

[0169] Referring to FIG. **11**A, in the state **1101**, the wearable device **101** that identifies a gaze facing a point P**2** within the screen **1112** may transmit information for displaying the screen **1112** to the external electronic device **120**, based on unfolding of the external electronic device **120**. The state **1103** of the external electronic device **120** of FIG. **11**A may be an exemplary state of the external electronic device **120** receiving the information for displaying the screen **1112**. According to an embodiment, the external electronic device **120** may bypass a screen for releasing a lock mode and display a screen **1130** corresponding to the screen **1112** by using the information received from the wearable device **101**. The external electronic device **120** may display the screen **1130** for continuously outputting media content included in the screen **1112**, by using the information. Referring to FIG. **11**A, in a state in which the wearable device **101** reproduces a specific timing (e.g., 3 minutes 35 seconds) of media content through the screen **1112**, in response to unfolding of the external electronic device **120**, the wearable device **101** may transmit information for displaying the media content to the external electronic device **120** according to the specific timing. The external electronic device **120** may output a portion of media content corresponding to the specific timing on the screen **1130**, by using the information.

[0170] Referring to FIG. **11**A, in the state **1101**, the wearable device **101** that identifies a gaze facing a point P**3** in the screen **1113** may transmit information for continuously executing an application corresponding to the screen **1113** to the external electronic device **120**, based on unfolding of the external electronic device **120**. A state **1104** of the external electronic device **120** of FIG. **11**A may be an exemplary state of the external electronic device **120** that receives the information for executing the application. When the application included in the information is installed in the external electronic device **120**, the external electronic device **120** may access a channel (e.g., a channel created for video conference) corresponding to the screen **1113** by executing the application. In the state **1104** in which the application included in the information is not installed in the external electronic device **120**, the external electronic device **120** may display a screen for installing the application. In the state **1104**, the external electronic device **120** may execute another application (e.g., a market application) capable of downloading the application included in the information. Based on completion of installation of the application, the external electronic device **120** may continuously display the screen **1113**.

[0171] According to an embodiment, when an application for voice call and/or video call is continuously executed, the wearable device **101** may perform call forwarding based on unfolding of the external electronic device **120**. Based on the call forwarding, the wearable device **101** may transmit information on a first call connection including the wearable device **101** to the unfolded external electronic device **120**. By using the information, the external electronic device **120** may execute the application to join the first call connection, or establish a second call connection related to the first call connection. By using the second call connection, the external electronic device **120** may support the user of the wearable device **101** to continuously perform voice call and/or video call by using the external electronic device **120**.

[0172] As in the exemplary state **1101** of FIG. **11**A, the wearable device **101** may selectively move a specific screen to the external electronic device **120** by using a gaze of the user **110** facing different screens **1111**, **1112**, and **1113**. The embodiment is not limited thereto, and the wearable device **101** may move a group of screens **1111**, **1112**, and **1113** to the external electronic device **120**. Referring to FIG. **11**B, an exemplary state **1106** in which the wearable device **101** displays a plurality of screens **1111**, **1112**, and **1114** on FoV of the user **110** is illustrated. The wearable device **101** may display a widget-type screen **1114** having a watch form based on execution of an application for displaying time. In the state **1106**, the wearable device **101** may receive an input for grouping the screens **1111**, **1112**, and **1114**. The input may include a gesture of connecting, dragging, and/or merging the screens **1111**, **1112**, and **1114**. In response to the input, the wearable device **101** may display a visual object **1160** indicating a group of screens **1111**, **1112**, and **1114**.

The visual object **1160** may be referred to as a wall in terms of having a form of a plane in which the screens **1111**, **1112**, and **1114** are arranged. Although the visual object **1160** having a rectangular plane form is exemplary illustrated, the embodiment is not limited thereto.

[0173] Referring to FIG. **11**B, in the state **1106** displaying the visual object **1160**, the wearable device **101** that identifies a gaze facing a point P**4** within the visual object **1160** may transmit information on the screens **1111**, **1112**, and **1114** grouped by the visual object **1160** to the external electronic device **120**, based on unfolding of the external electronic device **120**. The information may include data on a location, arrangement, and/or layout of the screens **1111**, **1112**, and **1114** within the visual object **1160**. The external electronic device **120** receiving the information may determine whether to display each of the screens **1111**, **1112**, and **1114** on the display, based on a threshold of the screens capable of displaying simultaneously on a display of the external electronic device **120**. In a state **1107** in which the threshold is 3 or more, the external electronic device **120** receiving information on the three screens **1111**, **1112**, and **1114** may execute applications corresponding to each of the screens **1111**, **1112**, and **1114**. Based on execution of the applications, the external electronic device **120** may display screens provided from the executed applications, similar to layout of the screens **1111**, **1112**, and **1114**.

[0174] Referring to FIG. **11**B, in a state **1108** in which the threshold value is less than **3**, the external electronic device **120** receiving information on the three screens **1111**, **1112**, and **1114** may display UI for identifying whether to display each of the screens **1111**, **1112**, and **1114**. In the state **1108** of FIG. **11**B, the external electronic device **120** may display visual objects **1181**, **1182**, and **1184** corresponding to each of the screens **1111**, **1112**, and **1114**. Visual objects **1181**, **1182**, and **1184** having a check box form are illustrated, but the embodiment is not limited thereto. In order to receive an input for completing selection of the screens **1111**, **1112**, and **1114**, the external electronic device **120** may display a visual object **1189** having a button form.

[0175] Referring to FIG. **11**B, in the state **1108** in which the visual objects **1182** and **1184** are selected, the external electronic device **120** may switch to a state **1109** in response to an input of selecting the visual object **1189**. In the state **1109**, the external electronic device **120** may selectively execute applications selected by the visual objects **1182** and **1184** among applications corresponding to the screens **1111**, **1112**, and **1114**. Referring to FIG. **11**B, the external electronic device **120** may display media content included in the screen **1112** based on the visual object **1182**, and may display a widget in a watch form corresponding to the screen **1114** by using the visual object **1184**.

[0176] Referring to FIG. **11**C, according to an embodiment, in a state **1190** displaying a plurality of screens **1111**, **1112**, and **1113**, based on unfolding of the external electronic device **120**, the wearable device **101** may display a visual object **1198** related to the unfolding. In the exemplary state **1190** of FIG. **11**C, the wearable device **101** may display a visual object **1198** having a pop-up window form including preset text (e.g., “phone opened”) representing unfolding of the external electronic device **120**. The embodiment is not limited thereto.

[0177] According to an embodiment, in the state **1190** identifying that unfolding of the external electronic device **120**, the wearable device **101** may display a visual object **1199** including different options related to at least one screen to be moved to the external electronic device **120** among the screens **1111**, **1112**, and **1113**. The options may be classified by a layout of at least one screen to be moved from the wearable device **101** to the external electronic device **120**. According to an embodiment, the wearable device **101** may obtain information on frequency in which each of the screens **1111**, **1112**, and **1113** is focused by the user **110**, based on a motion (e.g., gaze and/or hand gesture of the user **110**) of the user **110** wearing the wearable device **101**. By using the information, the wearable device **101** may transmit a signal for displaying at least one of the screens **1111**, **1112**, and **1113**, to the external electronic device **120**.

[0178] Referring to FIG. **11**C, according to an embodiment, buttons **1199-1**, **1199-2**, **1199-3**, and **1199-4** corresponding to each of options provided by the wearable device **101** to the user **110**

through the visual object **1199** are illustrated. In response to an input indicating to select button **1199-1**, the wearable device **101** may transmit a signal for displaying icons **1191-1** indicating each of the screens **1111**, **1112**, and **1113** displayed by the wearable device **101** to the external electronic device **120**. Based on the signal transmitted by an input indicating to select the button **1199-1**, the external electronic device **120** may switch to a state **1191**. In the state **1191**, the external electronic device **120** may display icons **1191-1** indicating applications corresponding to each of the screens **1111**, **1112**, and **1113** on a lower end (e.g., app tray) of a display.
[0179] Referring to FIG. **11**C, in response to an input indicating to select the button **1199-2** in the visual object **1199**, the wearable device **101** may transmit a signal to display a specific screen selected by information on the frequency of focusing by the user **110** in full screen, to the external electronic device **120**. The external electronic device **120** receiving the signal may display a screen **1192-1** corresponding to the signal, by switching to a state **1192**. For example, the wearable device **101** may transmit a signal for displaying the screen **1112** mostly focused by the user **110**, from among the screens **1111**, **1112**, and **1113** by using the information. In the state **1192** receiving the signal, the external electronic device **120** may display the screen **1192-1** on the entire displaying region of a display, by executing an application corresponding to the screen **1112**.
[0180] Referring to FIG. **11**C, in response to an input indicating to select a button **1199-3** in the visual object **1199**, the wearable device **101** may transmit a signal to display a specific screen selected by information on the frequency of focusing by the user **110** as a pop-up window on a display of the external electronic device **120**, to the external electronic device **120**. For example, the wearable device **101** may transmit a signal for displaying the screen **1112** mostly focused by the user **110**, from among the screens **1111**, **1112**, and **1113**, by using the information. Based on receiving the signal, the external electronic device **120** may switch to a state **1193**. In the state **1193**, the external electronic device **120** may display a pop-up window **1193-1** by overlapping on a screen displayed through a display, by executing an application corresponding to the screen **1112**.
[0181] Referring to FIG. **11**C, in response to an input indicating to select a button **1199-4** in the visual object **1199**, the wearable device **101** may transmit a signal for parallelly displaying the screens **1111**, **1112**, and **1113** displayed through the wearable device **101** to the external electronic device **120**. The signal may include a layout of the screens **1111**, **1112**, and **1113** arranged by information on the frequency of focusing by the user **110**. The layout may be set so that a sizes of the screens **1111**, **1112**, and **1113** increase in ascending order of the frequency in which the screens **1111**, **1112**, and **1113** are focused. Based on receiving the signal, the external electronic device **120** may switch to a state **1194**. In the state **1194**, the external electronic device **120** may display screens **1194-1**, **1194-2**, and **1194-3** having a layout corresponding to the signal on a display of the external electronic device **120**, based on the signal, by executing applications corresponding to each of the screens **1111**, **1112**, and **1113**. The external electronic device **120** may create screens **1194-1**, **1194-2**, and **1194-3** for continuously displaying each of the screens **1111**, **1112**, and **1113** displayed through the wearable device **101** based on the signal.
[0182] In the state **1190** of FIG. **11**C, the wearable device **101** may display the visual object **1199** during a time interval of a preset length. In the time interval displaying the visual object **1199**, in response to an input indicating to select at least one of the buttons **1199-1**, **1199-2**, **1199-3**, and **1199-4** in the visual object **1199**, the wearable device **101** may execute a function for moving at least one of the screens **1111**, **1112**, and **1113** to the external electronic device **120**. When any input related to the visual object **1199** is not identified during the time interval, the wearable device **101** may refrain from executing the function.
[0183] As described above, according to an embodiment, the wearable device **101** may execute one or more applications installed in the external electronic device **120**, by using a form of the external electronic device **120** and a direction of the gaze of the user **110** wearing the wearable device **101**. Based on execution of one or more of the applications, the wearable device **101** may move one or more screens (e.g., screens **1111**, **1112**, **1113**, and **1114**) floating on FoV of the user **110** to the

display of the external electronic device **120**.

[0184] An operation of the wearable device **101** performing a form of the external electronic device **120** based on whether it is a fully unfolded form (e.g., a form in which an angle of folding axis F of the external electronic device **120** is 180°) is described, but the embodiment is not limited thereto. For example, in another form distinguished from the fully unfolded form, such as the right angle form in Table 1, the wearable device **101** may control movement of the screen between the wearable device **101** and the external electronic device **120**.

[0185] Hereinafter, according to an embodiment, an example of an operation performed by the wearable device **101** based on identifying the right angle form of the external electronic device **120** will be described with reference to FIG. **12**.

[0186] FIG. **12** illustrates an example of an operation performed by a wearable device **101** by using a direction of an external electronic device **120**, according to an embodiment. The wearable device **101** and the external electronic device **120** of FIGS. **1** to **2** may include the wearable device **101** and the external electronic device **120** of FIG. **12**. An operation of the wearable device **101** described with reference to FIG. **12** may be related to at least one of the operations (e.g., the operation **340** of FIG. **3**) of FIG. **3**.

[0187] Referring to FIG. **12**, according to an embodiment, states **1201**, **1202**, and **1203** into which the wearable device **101** enters, according to a form of the external electronic device **120** foldable by a straight folding axis F are illustrated. In an exemplary case of FIG. **12**, the user **110** may lay down the external electronic device **120** or may arrange one surface of the external electronic device **120** toward the ground or a direction of gravity acceleration. For example, the folding axis F of the external electronic device **120** may be substantially perpendicular to the ground or the direction of gravity acceleration.

[0188] In the state **1201** identifying the fully folded external electronic device **120**, the wearable device **101** may display a screen **140** on a displaying region formed by a display (e.g., the display **220-1** of FIG. **2**). In the state **1201** in which the screen **140** for reproducing media content is displayed, the wearable device **101** may monitor a form of the external electronic device **120**. The wearable device **101** that identifies folding of the external electronic device **120** may switch from the state **1201** to the state **1202**. In the state **1202**, the wearable device **101** may display a visual object **1210** for guiding or informing of movement of the screen **140** based on an angle A**1** of folding axis F of the external electronic device **120**. Similar to the visual object **410** of FIG. **4**, the visual object **1210** of FIG. **12** may be moved, enlarged, and/or reduced according to the angle A**1** of the folding axis F. In an embodiment, the wearable device **101** may display the visual object **1210**, based on a gaze (e.g., a direction of a gaze facing the screen **140**) of the user **110** wearing the wearable device **101**, described above with reference to FIGS. **4** to **7**B. In an embodiment, the wearable device **101** may display the visual object **1210** based on a position of the external electronic device **120** in FoV of the user **110**, described with reference to FIGS. **8**A to **10**B.

[0189] In an embodiment, as in an exemplary case of FIG. **12**, when one surface of the external electronic device **120** is disposed toward a direction of gravity acceleration or the ground, the external electronic device **120** may be moved in an angle range (e.g., an angle range of 80° or more and 110° or less, in Table 1) including an angle A**1** of folding axis f that is in the right angle. When the angle A**1** is included in the angle range including the right angle, the external electronic device **120** may display UI suitable for a form of the display **220-2** classified by the folding axis F. According to an embodiment, the wearable device **101** may transmit information for displaying the screen **140** to the external electronic device **120**, based on whether the angle Al of folding axis F is included in the angle range in the state **1202**.

[0190] In the state **1202**, the wearable device **101** identifying the angle A**1** included in the angle range corresponding to the right angle form of Table 1 may enter the state **1203**. For example, in the state **1203**, an angle A**2** of folding axis F of the external electronic device **120** may be substantially 90°. The wearable device **101** may transmit information for continuously displaying the screen **140** to the external electronic device **120**, in response to identifying the angle A**2** included in the angle range including 90°. In the state **1203** of FIG. **12**, on the display **220-2** folded at a right angle by folding axis F, the external electronic device **120** may display a screen **1220** corresponding to the screen **140** on one surface that was positioned facing a direction of gravity acceleration or the ground and has been unfolded by the angle A**2**, in response to the information transmitted from the wearable device **101**. For example, the external electronic device **120** may display media content included in the screen **140** through the screen **1220**. Referring to FIG. **12**, similar to a play button **141** included in the screen **140**, the external electronic device **120** may display a play button **1221** for controlling reproduction of the media content on the screen **1220**.

[0191] Although an operation of the wearable device **101** is described based on the screen **140**, the embodiment is not limited thereto. For example, in the state **1203**, when the wearable device **101** moves a group of a plurality of screens to the external electronic device **120** as in the visual object **1160** of FIG. **11**B, the external electronic device **120** may display at least one of the plurality of screens, based on a form of the display **220-2** folded at a right angle by the folding axis F. In the example, a layout of the plurality of screens displayed through the display **220-2** of the external electronic device **120** may be different from a layout of a plurality of screens displayed by the wearable device **101**.

[0192] An operation of the wearable device **101** is described based on sequential movement of the states **1201**, **1202**, and **1203**, but the embodiment is not limited thereto. For example, in the state **1203** in which the external electronic device **120** has a right angle form, when the user **110** folds the external electronic device **120**, the wearable device **101** may execute a function for moving the screen **1220** of the display **220-2** to the wearable device **101**. The function may include a function of displaying a visual object (e.g., the visual object **510** of FIG. **5** and/or the visual object **840** of FIG. **8**B) for guiding or informing of movement of the screen **1220** similar as described above with reference to FIGS. **5** and/or **8**B.

[0193] As described above, according to an embodiment, the wearable device **101** may control movement of the screen **140** between the wearable device **101** and the external electronic device **120**, by using a form and/or direction of the external electronic device **120** having a straight folding axis F. The embodiment is not limited thereto. Hereinafter, an exemplary operation in which the wearable device **101** controls movement of the screen **140** will be described based on transformation of an external electronic device including a rollable display.

[0194] FIG. **13** illustrates an example of an operation performed by a wearable device **101** based on a form of an external electronic device **120**, according to an embodiment. The wearable device **101** of FIGS. **1** to **2** may include the wearable device **101** of FIG. **13**. An operation of the wearable device **101** described with reference to FIG. **13** may be related to at least one (e.g., operation **320** of FIG. **3**) of operations of FIG. **13**.

[0195] Referring to FIG. **13**, according to an embodiment, an exemplary state **1301** in which the wearable device **101** is connected to an external electronic device **120-3** is illustrated. The external electronic device **120-3** of FIG. **13** may include the external electronic device **120-3** of FIG. **2**. According to an embodiment, the wearable device **101** may identify an external electronic device **120-3** including a display **220**-insertable into a housing by using an image of a camera (e.g., the camera **225** of FIG. **2**). According to an embodiment, the wearable device **101** may identify the external electronic device **120-3** through a communication link established between the external electronic device **120** and the wearable device **101** by a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**).

[0196] Referring to FIG. **13**, according to an embodiment, a form of the external electronic device **120-3** connected to the wearable device **101** may be classified by a size of the display **220-2** inserted into a housing of the external electronic device **120-3**. For example, as the display **220-2** is inserted into the housing, a form, and/or a state of the external electronic device **120-3** in which a size of an externally exposed display **220-2** is minimized may be referred to as a rolled form (or a rolled state, fully rolled state). For example, as the display **220-2** is extracted from the housing, a form, and/or a state of the external electronic device **120-3** in which a size of an externally exposed display **220-2** is maximized may be referred to as an unrolled form (or an unrolled state, fully unrolled state). A form, and/or a state of the external electronic device **120-3** may include an intermediate form (or an intermediate state, partially rolled state, partially unrolled state) between the rolled form and the unrolled form.

[0197] Referring to FIG. **13**, in a state **1301** of identifying the external electronic device **120-3** in the rolled form, the wearable device **101** may display one or more screens **1111**, **1112**, and **1114** on a displaying region formed by a display (e.g., the display **220-1** of FIG. **2**). Referring to FIG. **13**, as in the state **1106** of FIG. **11**B, an exemplary state **1301** in which the wearable device **101** groups the screens **1111**, **1112**, and **1114** by using the visual object **1160** is illustrated. In the state **1301**, the wearable device **101** may identify deformation of the external electronic device **120-3** by using a camera and/or a communication circuit. The external electronic device **120-3** may be transformed into another form (e.g., unfolded form) different from the rolled form, in response to an Input (e.g., input of pressing a button of the external electronic device **120-3**) indicating external force (e.g., external force applied to the external electronic device **120-3** by the user **110**) and/or deformation. Deformation of the external electronic device **120-3** may be performed by an actuator included in the external electronic device **120-3**.

[0198] In the state **1301** of FIG. **13**, the wearable device **101** that identifies deformation of the external electronic device **120-3** may enter a transition mode. For example, the wearable device **101** may enter the transition mode, based on a size of the display **220-2** extracted from a housing of the external electronic device **120-3**. For example, the wearable device **101** that identifies the display **220-2** extracted by exceeding the preset size may enter the transition mode. In the transition mode, the wearable device **101** may display a visual object (e.g., the visual object **410** of FIG. **4**) for guiding or informing of movement of at least one screen.

[0199] Referring to FIG. **13**, in the transition mode, the wearable device **101** may select at least one screen to be moved to the external electronic device **120-3** by using a direction of a gaze of the user **110**. For example, the wearable device **101** that identifies a gaze facing a point P**1** in the screen **1111** may transmit information for moving the screen **1111** to the external electronic device **120-3**, in response to extraction of the display **220-2** of the external electronic device **120-3**. The state **1302** of the external electronic device **120-3** of FIG. **13** may be an exemplary state of the external electronic device **120-3** that receives information for moving the screen **1111**. As the external electronic device **120-3** is transformed into unfolded form, the external electronic device **120-3** may execute an application corresponding to the screen **1111** by using the information and display a screen provided from the application on the display **220-2**. As described above with reference to FIG. **11**A, the external electronic device **120-3** may release a lock mode by displaying a screen related to the lock mode or using a user account logged in to the wearable device **101**.

[0200] Referring to FIG. **13**, the wearable device **101** that identifies a gaze facing a point P**2** in the visual object **1160** may transmit information on the screens **1111**, **1112**, and **1114** grouped by the visual object **1160** to the external electronic device **120-3**. The information may include data on one or more applications that provide each of the screens **1111**, **1112**, and **1114**, and/or a layout of the screens **1111**, **1112**, and **1114**. The external electronic device **120-3** receiving the information may display at least one of the screens **1111**, **1112**, and **1114**, based on a threshold of screens capable of being displayed simultaneously on the display **220-2**. When the wearable device **101** transmits information on the three screens **1111**, **1112**, and **1114**, and the threshold is less than **3**, the external electronic device **120-3** may display UI for selection of the screens **1111**, **1112**, and **1114**, as in the state **1303**. Referring to FIG. **13**, in the state **1303**, the external electronic device **120-3** may display visual objects **1321**, **1322**, and **1323** corresponding to each of the screens **1111**, **1112**, and **1114**. Visual objects **1321**, **1322**, and **1323** having a check box form are illustrated, but embodiments are not limited thereto. In order to receive an input for completing selection of the screens **1111**, **1112**, and **1114**, the external electronic device **120-3** may display a visual object **1329** having a button form.

[0201] Referring to FIG. **13**, in the state **1303** in which the visual objects **1321** and **1322** are selected, the external electronic device **120-3** may switch to the state **1304**, in response to an input with respect to the visual object **1329**. In the state **1304**, the external electronic device **120-3** may execute one or more applications corresponding to the screens **1112** and **1114** selected by the visual objects **1321** and **1322**. For example, the external electronic device **120-3** may display media content included in the screen **1112** on the display **220-2**, and display a watch form widget corresponding to the screen **1114**.

[0202] Although an operation of the wearable device **101** based on a direction of a gaze facing any one of the points P**1** and P**2** is described, the embodiment is not limited thereto. For example, as described above with reference to FIG. **8**A, the wearable device **101** may select one or more screens to be moved from the wearable device **101** to the external electronic device **120-3**, based on the external electronic device **120-3** at least partially overlapping with any one of the screens **1111**, **1112**, and **1114**, and/or the visual object **1160**.

[0203] An operation of the wearable device **101** is described based on an exemplary case in which the display **220-2** of the external electronic device **120-3** is extracted, but the embodiment is not limited thereto. For example, in response to identifying that the external electronic device **120-3** of the state **1304** is transformed into a rolled form based on insertion of the display **220-2**, the wearable device **101** may execute a function for displaying at least one screen displayed on the display **220-2** of the external electronic device **120-3** by using a display of the wearable device **101**. As described above with reference to FIG. **5**, the function may be executed according to a direction of a gaze of the user **110**. The embodiment is not limited thereto, and the function may be executed based on a location of the external electronic device **120-3** within FoV of the user **110**, as described above with reference to FIG. **8**B.

[0204] As described above, according to an embodiment, the wearable device **101** may control movement of a screen between the wearable device **101** and the external electronic device **120**, based on deformation of foldable or rollable external electronic device **120**. When a size of a display (e.g., the display **220-2** of FIG. **2**) of the external electronic device **120** viewed by the user **110** increases based on the transformation, the wearable device **101** may move information displayed through a display (e.g., the display **220-1** of FIG. **2**) of the wearable device **101** to the display with an increased size. When a size of the display of the external electronic device **120** viewed by the user **110** is reduced, the wearable device **101** may move information displayed through the display of the external electronic device **120** to the display of the wearable device **101**. Based on movement of information described above, the wearable device **101** may support seamless movement between the wearable device **101** and the external electronic device **120**.

[0205] Hereinafter, according to an embodiment, an example of a form factor of the wearable device **101** will be described by using FIGS. **14**A to **14**B and/or **15**A to **15**B.

[0206] FIG. **14**A illustrates an example of a perspective view of a wearable device, according to an embodiment. FIG. **14**B illustrates an example of one or more hardware disposed in a wearable device, according to an embodiment. The wearable device **101** of FIGS. **1** to **2** may include a wearable device **1400** of FIGS. **14**A to **14**B. As shown in FIG. **14**A, according to an embodiment, the wearable device **1400** may include at least one display **1450** and a frame supporting the at least one display **1450**.

[0207] According to an embodiment, the wearable device **1400** may be wearable on a part of the user's body. The wearable device **1400** may provide augmented reality (AR), virtual reality (VR), or mixed reality (MR) combining the augmented reality and the virtual reality to a user wearing the wearable device **1400**. For example, the wearable device **1400** may output a virtual reality image to a user through the at least one display **1450** in response to a user's preset gesture obtained through a motion recognition camera **1440-2** of FIG. **14**B.

[0208] According to an embodiment, the at least one display **1450** in the wearable device **1400** may provide visual information to a user. The at least one display **1450** may include the display **220-1** of FIG. **2**. For example, the at least one display **1450** may include a transparent or translucent lens. The at least one display **1450** may include a first display **1450-1** and/or a second display **1450-2** spaced apart from the first display **1450-1**. For example, the first display **1450-1** and the second display **1450-2** may be disposed at positions corresponding to the user's left and right eyes, respectively.
[0209] Referring to FIG. **14**B, the at least one display **1450** may provide another visual information, which is distinct from the visual information, together with the visual information included in the ambient light passing through the lens, a user wearing the wearable device **1400**, by forming a displaying region on the lens. The lens may be formed based on at least one of a fresnel lens, a pancake lens, or a multi-channel lens. For example, the displaying region formed by the at least one display **1450** may be formed on the second surface **1432** among the first surface **1431** and the second surface **1432** of the lens. When the user wears the wearable device **1400**, the ambient light may be transmitted to the user by being incident on the first surface **1431** and being penetrated through the second surface **1432**. For another example, the at least one display **1450** may display the virtual reality image to be combined with a real screen transmitted through the ambient light. The virtual reality image outputted from the at least one display **1450** may be transmitted to the user's eyes through the one or more hardware (e.g., optical devices **1482** and **1484**, and/or at least one waveguides **1433** and **1434**)) included in the wearable device **1400**.
[0210] According to an embodiment, the wearable device **1400** may include the waveguides **1433** and **1434** that diffracts light transmitted from the at least one display **1450** and relayed by the optical devices **1482** and **1484** and transmits it to the user. The waveguides **1433** and **1434** may be formed based on at least one of glass, plastic, or polymer. A nano pattern may be formed on at least a portion of the outside or inside of the waveguides **1433** and **1434**. The nano pattern may be formed based on a grating structure having a polygonal or curved shape. Light incident to one end of the waveguides **1433** and **1434** may be propagated to the other end of the waveguides **1433** and **1434** by the nano pattern. The waveguides **1433** and **1434** may include at least one of at least one diffraction element (e.g., a diffractive optical element (DOE), a holographic optical element (HOE)), and a reflection element (e.g., a reflection mirror). For example, the waveguides **1433** and **1434** may be disposed in the wearable device **1400** to guide a screen displayed by the at least one display **1450** to the user's eyes. For example, the screen may be transmitted to the user's eyes through total internal reflection (TIR) generated in the waveguides **1433** and **1434**.
[0211] According to an embodiment, the wearable device **1400** may analyze an object included in a real image collected through a photographing camera **1440-1**, combine virtual object corresponding to an object that become a subject of augmented reality provision among the analyzed object, and display them on the at least one display **1450**. The virtual object may include at least one of text and images for various information associated with the object included in the real image. The wearable device **1400** may analyze the object by using a multi-camera such as a stereo camera. For the object analysis, the wearable device **1400** may execute time-of-flight (ToF) and/or simultaneous localization and mapping (SLAM) supported by the multi-camera. The user wearing the wearable device **1400** may watch an image displayed on the at least one display **1450**.
[0212] According to an embodiment, the frame may be configured with a physical structure in which the wearable device **1400** may be worn on the user's body. According to an embodiment, the frame may be configured so that when the user wears the wearable device **1400**, the first display **1450-1** and the second display **1450-2** may be positioned corresponding to the user's left and right eyes. The frame may support the at least one display **1450**. For example, the frame may support the first display **1450-1** and the second display **1450-2** to be positioned at positions corresponding to the user's left and right eyes.
[0213] Referring to FIG. **14**A, according to an embodiment, the frame may include a region **1420**

at least partially in contact with the portion of the user's body in case that the user wears the wearable device **1400**. For example, the region **1420** in contact with the portion of the user's body of the frame may include a region contacting a portion of the user's nose, a portion of the user's ear, and a portion of the side of the user's face that the wearable device **1400** contacts. According to an embodiment, the frame may include a nose pad **1410** that is contacted on the portion of the user's body. When the wearable device **1400** is worn by the user, the nose pad **1410** may be contacted on the portion of the user's nose. The frame may include a first temple **1404** and a second temple **1405** that is contacted on another portion of the user's body that is distinct from the portion of the user's body.

[0214] For example, the frame may include a first rim **1401** surrounding at least a portion of the first display **1450-1**, a second rim **1402** surrounding at least a portion of the second display **1450-2**, a bridge **1403** disposed between the first rim **1401** and the second rim **1402**, a first pad **1411** disposed along a portion of the edge of the first rim **1401** from one end of the bridge **1403**, a second pad **1412** disposed along a portion of the edge of the second rim **1402** from the other end of the bridge **1403**, the first temple **1404** extending from the first rim **1401** and fixed to a portion of the wearer's ear, and the second temple **1405** extending from the second rim **1402** and fixed to a portion of the ear opposite to the ear. The first pad **1411** and the second pad **1412** may be in contact with the portion of the user's nose, and the first temple **1404** and the second temple **1405** may be in contact with a portion of the user's face and the portion of the user's ear. The temples **1404** and **1405** may be rotatably connected to the rim through hinge units **1406** and **1407** of FIG. **14**B. The first temple **1404** may be rotatably connected with respect to the first rim **1401** through the first hinge unit **1406** disposed between the first rim **1401** and the first temple **1404**. The second temple **1405** may be rotatably connected with respect to the second rim **1402** through the second hinge unit **1407** disposed between the second rim **1402** and the second temple **1405**. According to an embodiment, the wearable device **1400** may identify an external object (e.g., a user's fingertip) touching the frame and/or a gesture performed by the external object by using a touch sensor, a grip sensor, and/or a proximity sensor formed on at least a portion of the surface of the frame.

[0215] According to an embodiment, the wearable device **1400** may include hardware (e.g., hardware described above based on the block diagram of FIG. **2**) that performs various functions. For example, the hardware may include a battery module **1470**, an antenna module **1475**, the optical devices **1482** and **1484**, speakers **1492-1** and **1492-2**, microphones **1494-1**, **1494-2**, and **1494-3**, a light emitting module (not illustrated), and/or a printed circuit board **1490**. Various hardware may be disposed in the frame.

[0216] According to an embodiment, the microphone **1494-1**, **1494-2**, and **1494-3** of the wearable device **1400** may obtain a sound signal, by being disposed on at least a portion of the frame. The first microphone **1494-1** disposed on the nose pad **1410**, the second microphone **1494-2** disposed on the second rim **1402**, and the third microphone **1494-3** disposed on the first rim **1401** are illustrated in FIG. **14**B, but the number and disposition of the microphone **1494** are not limited to an embodiment of FIG. **14**B. In case that the number of the microphone **1494** included in the wearable device **1400** is two or more, the wearable device **1400** may identify the direction of the sound signal by using a plurality of microphones disposed on different portions of the frame.

[0217] According to an embodiment, the optical devices **1482** and **1484** may transmit the virtual object transmitted from the at least one display **1450** to the waveguides **1433** and **1434**. For example, the optical devices **1482** and **1484** may be a projector. The optical devices **1482** and **1484** may be disposed adjacent to the at least one display **1450** or may be included in the at least one display **1450** as portion of the at least one display **1450**. The first optical device **1482** may correspond to the first display **1450-1**, and the second optical device **1484** may correspond to the second display **1450-2**. The first optical device **1482** may transmit the light outputted from the first display **1450-1** to the first waveguide **1433**, and the second optical device **1484** may transmit light outputted from the second display **1450-2** to the second waveguide **1434**.

[0218] In an embodiment, a camera **1440** may include an eye tracking camera (ET CAM) **1440-1**, the motion recognition camera **1440-2**, and/or the photographing camera **1440-3**. The photographing camera **1440-3**, the eye tracking camera **1440-1**, and the motion recognition camera **1440-2** may be disposed at different positions on the frame and may perform different functions. The photographing camera **1440-3**, the eye tracking camera **1440-1**, and the motion recognition camera **1440-2** may be an example of the camera **225** of FIG. **2**. The eye tracking camera **1440-1** may output data indicating the gaze of the user wearing the wearable device **1400**. For example, the wearable device **1400** may detect the gaze from an image including the user's pupil obtained through the eye tracking camera **1440-1**. An example in which the eye tracking camera **1440-1** is disposed toward the user's right eye is illustrated in FIG. **14**B, but the embodiment is not limited thereto, and the eye tracking camera **1440-1** may be disposed alone toward the user's left eye or may be disposed toward two eyes.

[0219] In an embodiment, the photographing camera **1440-3** may photograph a real image or background to be matched with a virtual image in order to implement the augmented reality or mixed reality content. The photographing camera may photograph an image of a specific object existing at a position viewed by the user and may provide the image to the at least one display **1450**. The at least one display **1450** may display one image in which a virtual image provided through the optical devices **1482** and **1484** is overlapped with information on the real image or background including an image of the specific object obtained by using the photographing camera. In an embodiment, the photographing camera may be disposed on the bridge **1403** disposed between the first rim **1401** and the second rim **1402**.

[0220] In an embodiment, the eye tracking camera **1440-1** may implement a more realistic augmented reality by matching the user's gaze with the visual information provided on the at least one display **1450** by tracking the gaze of the user wearing the wearable device **1400**. For example, when the user looks at the front, the wearable device **1400** may naturally display environment information associated with the user's front on the at least one display **1450** at the position where the user is positioned. The eye tracking camera **1440-1** may be configured to capture an image of the user's pupil in order to determine the user's gaze. For example, the eye tracking camera **1440-1** may receive gaze detection light reflected from the user's pupil and may track the user's gaze by using the position and movement of the received gaze detection light. In an embodiment, the eye tracking camera **1440-1** may be disposed at a position corresponding to the user's left and right eyes. For example, the eye tracking camera **1440-1** may be disposed in the first rim **1401** and/or the second rim **1402** to face the direction in which the user wearing the wearable device **1400** is positioned.

[0221] In an embodiment, the motion recognition camera **1440-2** may provide a specific event to the screen provided on the at least one display **1450** by recognizing the movement of the whole or portion of the user's body, such as the user's torso, hand, or face. The motion recognition camera **1440-2** may obtain a signal corresponding to the gesture by recognizing the user's gesture, and may provide a display corresponding to the signal to the at least one display **1450**. The processor may identify a signal corresponding to the operation and may perform a preset function based on the identification. In an embodiment, the motion recognition camera **1440-2** may be disposed on the first rim **1401** and/or the second rim **1402**.

[0222] In an embodiment, the camera **1440** included in the wearable device **1400** is not limited to the above-described eye tracking camera **1440-1** and the motion recognition camera **1440-2**. For example, the wearable device **1400** may identify an external object included in the FoV by using the photographing camera **1440-3** disposed toward the user's FoV. That the wearable device **1400** identifies the external object may be performed by using a sensor for identifying a distance between the wearable device **1400** and the external object, such as a depth sensor and/or a time of flight (ToF) sensor. The camera **1440** disposed toward the FoV may support an autofocus function and/or an optical image stabilization (OIS) function. For example, the wearable device **1400** may include the camera **1440** (e.g., a face tracking (FT) camera) disposed toward the face in order to obtain an image including the face of the user wearing the wearable device **1400**.

[0223] Although not illustrated, the wearable device **1400** according to an embodiment may further include a light source (e.g., LED) that emits light toward a subject (e.g., user's eyes, face, and/or an external object in the FoV) photographed by using the camera **1440**. The light source may include an LED having an infrared wavelength. The light source may be disposed on at least one of the frame, and the hinge units **1406** and **1407**.

[0224] According to an embodiment, the battery module **1470** may supply power to electronic components of the wearable device **1400**. In an embodiment, the battery module **1470** may be disposed in the first temple **1404** and/or the second temple **1405**. For example, the battery module **1470** may be a plurality of battery modules **1470**. The plurality of battery modules **1470**, respectively, may be disposed on each of the first temple **1404** and the second temple **1405**. In an embodiment, the battery module **1470** may be disposed at an end of the first temple **1404** and/or the second temple **1405**.

[0225] In an embodiment, the antenna module **1475** may transmit the signal or power to the outside of the wearable device **1400** or may receive the signal or power from the outside. The antenna module **1475** may be electronically and/or operably connected to a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**) of the wearable device **1400**. In an embodiment, the antenna module **1475** may be disposed in the first temple **1404** and/or the second temple **1405**. For example, the antenna module **1475** may be disposed close to one surface of the first temple **1404** and/or the second temple **1405**.

[0226] In an embodiment, the speakers **1492-1** and **1492-2** may output a sound signal to the outside of the wearable device **1400**. A sound output module may be referred to as a speaker. In an embodiment, the speakers **1492-1** and **1492-2** may be disposed in the first temple **1404** and/or the second temple **1405** in order to be disposed adjacent to the ear of the user wearing the wearable device **1400**. For example, the wearable device **1400** may include the second speaker **1492-2** disposed adjacent to the user's left ear by being disposed in the first temple **1404**, and the first speaker **1492-1** disposed adjacent to the user's right ear by being disposed in the second temple **1405**.

[0227] In an embodiment, the light emitting module (not illustrated) may include at least one light emitting element. The light emitting module may emit light of a color corresponding to a specific state or may emit light through an operation corresponding to the specific state in order to visually provide information on a specific state of the wearable device **1400** to the user. For example, in case that the wearable device **1400** needs charging, it may repeatedly emit red light at a preset timing. In an embodiment, the light emitting module may be disposed on the first rim **1401** and/or the second rim **1402**.

[0228] Referring to FIG. **14**B, according to an embodiment, the wearable device **1400** may include the printed circuit board (PCB) **1490**. The PCB **1490** may be included in at least one of the first temple **1404** or the second temple **1405**. The PCB **1490** may include an interposer disposed between at least two sub PCBs. On the PCB **1490**, one or more hardware (e.g., hardware illustrated by the blocks described above with reference to FIG. **2**) included in the wearable device **1400** may be disposed. The wearable device **1400** may include a flexible PCB (FPCB) for interconnecting the hardware.

[0229] According to an embodiment, the wearable device **1400** may include at least one of a gyro sensor, a gravity sensor, and/or an acceleration sensor for detecting the posture of the wearable device **1400** and/or the posture of a body part (e.g., a head) of the user wearing the wearable device **1400**. Each of the gravity sensor and the acceleration sensor may measure gravity acceleration, and/or acceleration based on preset 3-dimensional axes (e.g., x-axis, y-axis, and z-axis) perpendicular to each other. The gyro sensor may measure angular velocity of each of preset 3-dimensional axes (e.g., x-axis, y-axis, and z-axis). At least one of the gravity sensor, the acceleration sensor, and the gyro sensor may be referred to as an inertial measurement unit (IMU). According to an embodiment, the wearable device **1400** may identify the user's motion and/or gesture performed to execute or stop a specific function of the wearable device **1400** by using the IMU.

[0230] FIGS. **15**A to **15**B illustrate an example of an exterior of a wearable device **1500**, according to an embodiment. The wearable device **101** of FIGS. **1** to **2** may include the wearable device **1500** of FIGS. **15**A to **15**B. According to an embodiment, an example of an exterior of a first surface **1510** of a housing of the wearable device **1500** may be illustrated in FIG. **15**A, and an example of an exterior of a second surface **1520** opposite to the first surface **1510** may be illustrated in FIG. **15**B.

[0231] Referring to FIG. **15**A, according to an embodiment, the first surface **1510** of the wearable device **1500** may have an attachable shape on the user's body part (e.g., the user's face). Although not illustrated, the wearable device **1500** may further include a strap for being fixed on the user's body part, and/or one or more temples (e.g., the first temple **1404** and/or the second temple **1405** of FIGS. **14**A to **14**B). A first display **1550-1** for outputting an image to the left eye among the user's two eyes and a second display **1550-2** for outputting an image to the right eye among the user's two eyes may be disposed on the first surface **1510**. The wearable device **1500** may be formed on the first surface **1510** and may further include rubber or silicon packing for preventing interference by light (e.g., ambient light) different from the light emitted from the first display **1550-1** and the second display **1550-2**.

[0232] According to an embodiment, the wearable device **1500** may include cameras **1540-3** and **1540-4** for photographing and/or tracking two eyes of the user adjacent to each of the first display **1550-1** and the second display **1550-2**. The cameras **1540-3** and **1540-4** may be referred to as ET cameras. According to an embodiment, the wearable device **1500** may include cameras **1540-1** and **1540-2** for photographing and/or recognizing the user's face. The cameras **1540-1** and **1540-2** may be referred to as FT cameras.

[0233] Referring to FIG. **15**B, a camera (e.g., cameras **1540-5**, **1540-6**, **1540-7**, **1540-8**, **1540-9**, and **1540-10**), and/or a sensor (e.g., the depth sensor **1530**) for obtaining information associated with the external environment of the wearable device **1500** may be disposed on the second surface **1520** opposite to the first surface **1510** of FIG. **15**A. For example, the cameras **1540-5**, **1540-6**, **1540-7**, **1540-8**, **1540-9**, and **1540-10** may be disposed on the second surface **1520** in order to recognize an external object different from the wearable device **1500**. For example, by using cameras **1540-9**, and **1540-10**, the wearable device **1500** may obtain an image and/or video to be transmitted to each of the user's two eyes. The camera **1540-9** may be disposed on the second surface **1520** of the wearable device **1500** to obtain an image to be displayed through the second display **1550-2** corresponding to the right eye among the two eyes. The camera **1540-10** may be disposed on the second surface **1520** of the wearable device **1500** to obtain an image to be displayed through the first display **1550-1** corresponding to the left eye among the two eyes.

[0234] According to an embodiment, the wearable device **1500** may include the depth sensor **1530** disposed on the second surface **1520** in order to identify a distance between the wearable device **1500** and the external object. By using the depth sensor **1530**, the wearable device **1500** may obtain spatial information (e.g., a depth map) about at least a portion of the FoV of the user wearing the wearable device **1500**.

[0235] Although not illustrated, a microphone for obtaining sound outputted from the external object may be disposed on the second surface **1520** of the wearable device **1500**. The number of microphones may be one or more according to embodiments.

[0236] As described above, according to an embodiment, the wearable device **1500** may have a form factor for being worn on the user's head. The wearable device **1500** in a state worn on the head may provide a user experience based on augmented reality and/or mixed reality. The wearable device **1500** may display UI including an external electronic device (e.g., the external electronic device **120** of FIG. **1**) by using the first display **1550-1** and the second display **1550-2**. A user wearing the wearable device **1500** may view media content displayed by the wearable device **1500** together with the external electronic device through the UI. The wearable device **1500** may execute a function for moving the media content to the external electronic device based on deformation of the external electronic device.

[0237] In an embodiment, a method for the wearable device to move information between the wearable device and the external electronic device may be required based on deformation of the external electronic device. As described above, according to an embodiment, a wearable device (e.g., the wearable device **101** of FIGS. **1** to **2**) may comprise a communication circuitry (e.g., the communication circuit **240-1** of FIG. **2**), a display (e.g., the display **220-1** of FIG. **2**), memory (e.g., the memory **215-1** of FIG. **2**) storing instructions, and a processor (e.g., the processor **210-1** of FIG. **2**). The instructions, when executed by the processor, may cause the wearable device to obtain information with respect to a form of an external electronic device (e.g., the external electronic device **120** of FIGS. **1** to **2**). The instructions, when executed by the processor, may cause the wearable device to, in response to identifying that the external electronic device is transformed into a first form based on the information, execute a function to move at least one first screen of the display to a display of the external electronic device. The instructions, when executed by the processor, may cause the wearable device to, in response to identifying that the external electronic device is transformed into a second form different from the first form based on the information, display through the display at least one second screen provided by the external electronic device. According to an embodiment, the wearable device may move at least one screen between the wearable device and the external electronic device based on deformation of the external electronic device.

[0238] For example, the instructions, when executed by the processor, may cause the wearable device to, in a state identifying the external electronic device including a flexible display foldable by a folding axis (e.g., the folding axis F of FIG. **1**), identify a form of the external electronic device as being the first form or the second form based on an angle that the flexible display is folded by the folding axis.

[0239] For example, the wearable device may further comprise a camera (e.g., the camera **225** of FIG. **2**). The instructions, when executed by the processor, may cause the wearable device to identify the angle by which the flexible display is folded with respect to the folding axis, based on identifying the flexible display by using an image of the camera.

[0240] For example, the instructions, when executed by the processor, may cause the wearable device to obtain the information including the angle through the communication circuitry.

[0241] For example, the instructions, when executed by the processor, may cause the wearable device to, while the external electronic device is transformed into the first form, display a visual object (e.g., the visual object **410** of FIG. **4**, the visual object **810** of FIG. **8**A, and/or the visual object **1210** of FIG. **12**) for guiding or informing that the at least one first screen is moved to the flexible display of the external electronic device.

[0242] For example, the instructions, when executed by the processor, may cause the wearable device to, while the external electronic device is transformed from the second form into an intermediate form between the first form and the second form, display a visual object (e.g., the visual object **510** of FIG. **5**, the visual object **840** of FIG. **8**B) for guiding or informing that the at least one second screen is moved to the display.

[0243] For example, the instructions, when executed by the processor, may cause the wearable device to identify that the external electronic device is transformed into the first form, based on whether the angle of the flexible display folded by the folding axis exceeds to a first threshold angle. The instructions, when executed by the processor, may cause the wearable device to identify that the external electronic device is transformed into the second form, based on whether the angle of the flexible display folded by the folding axis is reduced lower or smaller than a second

threshold angle smaller than the first threshold angle.
[0244] For example, the instructions, when executed by the processor, may cause the wearable device to cease to display the at least one first screen on the display, in response to transmitting a first signal to the external electronic device to display the at least one first screen through a display of the external electronic device.
[0245] For example, the wearable device may further comprise a camera. The instructions, when executed by the processor, may cause the wearable device to select the at least one first screen corresponding to the direction, among a plurality of screens executed using the display, based on identifying a direction of gaze from an image of the camera.
[0246] For example, the instructions, when executed by the processor, may cause the wearable device to, in a state transmitting a first signal to display a plurality of screens through the external electronic device, transmit the first signal to the external electronic device which includes information with respect to a layout of the plurality of screens.
[0247] For example, the instructions, when executed by the processor, may cause the wearable device to display a visual object for guiding or informing of movement of the at least one second screen on the display, in response to receiving a third signal indicating the at least one second screen from the external electronic device after transmitting a second signal to display the at least one second screen through the display of the wearable device.
[0248] For example, the instructions, when executed by the processor, may cause the wearable device to, in a state identifying the external electronic device including a flexible display insertable into a housing, identify a form of the external electronic device as being the first form or the second form based on a size that the flexible display is extracted from the size.
[0249] For example, the instructions, when executed by the processor, may cause the wearable device to, in a state in which the external electronic device is viewed through a preset portion within the display, identify transformation of the external electronic device by using the information.
[0250] As described above, according to an embodiment, a method of a wearable device may comprise obtaining information with respect to a form of an external electronic device. The method may comprise, in response to identifying that the external electronic device is transformed into a first form based on the information, executing a function to move at least one first screen to a display of the external electronic device. The method may comprise, in response to identifying that the external electronic device is transformed into a second form different from the first form based on the information, displaying through the display at least one second screen provided by the external electronic device.
[0251] For example, the obtaining may comprise, in a state identifying the external electronic device including a flexible display foldable by a folding axis, identifying a form of the external electronic device as being the first form or the second form based on an angle that the flexible display is folded by the folding axis.
[0252] For example, the obtaining may comprise, based on identifying the flexible display by using an image of a camera of the wearable device, identify the angle that the flexible display is folded by the folding axis.
[0253] For example, the obtaining may comprise obtain the information including the angle through the communication circuitry.
[0254] For example, the transmitting the first signal may, while the external electronic device is transformed into the first form, display a visual object for guiding or informing that the at least one first screen is moved to the flexible display of the external electronic device.
[0255] As described above, according to an embodiment, a wearable device (e.g., the wearable device **101** of FIGS. **1** to **2**) may comprise a camera (e.g., the camera **225** of FIG. **2**), a communication circuit (e.g., the communication circuit **240-1** of FIG. **2**), a display (e.g., the display **220-1** of FIG. **2**), memory (e.g., the memory **215-1** of FIG. **2**) storing instructions, and a processor

(e.g., the processor **210-1** of FIG. **2**). The instructions, when executed by the processor, may cause the wearable device to obtain information on a form of the external electronic device, in a state in which an external electronic device (e.g., the external electronic device **120** of FIGS. **1** to **2**) viewed through the display by using the camera is identified. The instructions, when executed by the processor, may cause the wearable device to display a visual object (e.g., the visual object **410** of FIG. **4**, the visual object **510** of FIG. **5**, the visual object **810** of FIG. **8**A, the visual object **840** of FIG. **8**B, and/or the visual object **1210** of FIG. **12**) on the display to guide the movement of at least one media content (e.g., the media content **280** of FIG. **2**), based on identifying transformation of a flexible display of the external electronic device by using the information. The instructions, when executed by the processor, may cause the wearable device to, while the visual object is displayed, execute a function for moving at least one media content between the wearable device and the external electronic device, based on transformation of the flexible display viewed through the display.

[0256] For example, the instructions, when executed by the processor, may cause the wearable device to transmit a signal for displaying the at least one media content by using the flexible display to the external electronic device through the communication circuit, based on identifying unfolding of the flexible display by using the information.

[0257] For example, the instructions, when executed by the processor, may cause the wearable device to display the visual object moving toward the flexible display, based on identifying the flexible display that is unfolded by exceeding a preset angle.

[0258] For example, the instructions, when executed by the processor, may cause the wearable device to display the visual object moving from the flexible display, based on identifying the flexible display that is folded below the preset angle.

[0259] For example, the instructions, when executed by the processor, may cause the wearable device to display the visual object moving toward the flexible display, based on identifying the flexible display that is extracted from a housing of the external electronic device.

[0260] For example, the instructions, when executed by the processor, may cause the wearable device to display the visual object moving from the flexible display, based on identifying the flexible display that is inserted into a housing of the external electronic device.

[0261] As described above, according to an embodiment, a method of a wearable device may comprise obtaining information on a form of the external electronic device, in a state in which an external electronic device viewed through a display of the wearable device by using a camera of the wearable device is identified. The method may comprise displaying a visual object on the display to guide the movement of at least one media content, based on identifying transformation of a flexible display of the external electronic device by using the information. The method may comprise, while the visual object is displayed, executing a function for moving at least one media content between the wearable device and the external electronic device, based on transformation of the flexible display viewed through the display.

[0262] For example, the executing may comprise transmitting a signal for displaying the at least one media content by using the flexible display to the external electronic device through the communication circuit, based on identifying unfolding of the flexible display by using the information.

[0263] For example, the executing may comprise displaying the visual object moving toward the flexible display, based on identifying the flexible display that is unfolded by exceeding a preset angle.

[0264] For example, the executing may comprise displaying the visual object moving from the flexible display, based on identifying the flexible display that is folded below the preset angle.

[0265] For example, the executing may comprise displaying the visual object moving toward the flexible display, based on identifying the flexible display that is extracted from a housing of the external electronic device.

[0266] For example, the executing may comprise displaying the visual object moving from the flexible display, based on identifying the flexible display that is inserted into a housing of the external electronic device.
[0267] According to an embodiment, a wearable device may comprise a communication circuitry, a display, memory configured to store instructions, and a processor. The instructions, when executed by the processor, cause the wearable device to display within the display at least one first screen. The instructions, when executed by the processor, cause the wearable device to transmit, in response to identifying that an external electronic device is transformed into a first state based on the information, to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device. The instructions, when executed by the processor, cause the wearable device to transmit, in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, information associated with at least one third screen displayed through the external electronic device. The instructions, when executed by the processor, cause the wearable device to display within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device.
[0268] For example, the instructions, when executed by the processor, cause the wearable device to identify the external electronic device including a hinge structure of a plurality of housings. The instructions, when executed by the processor, cause the wearable device to identify the state of the external electronic device as being the first state or the second state based on an angle between housings of the external electronic device.
[0269] For example, the wearable device may comprise a first camera. The instructions, when executed by the processor, cause the wearable device to based on identifying the flexible display by using an image captured by the first camera, identify the angle by which the flexible display is folded with respect to the folding axis.
[0270] For example, the instructions, when executed by the processor, cause the wearable device to obtain the information including the angle through the communication circuitry.
[0271] For example, the instructions, when executed by the processor, cause the wearable device to display, while the external electronic device is transformed into the first state, a visual object for guiding that the at least one first screen is moved to the flexible display of the external electronic device.
[0272] For example, the instructions, when executed by the processor, cause the wearable device to display, while the external electronic device is transformed from the second state into an intermediate state between the first state and the second state, a visual object for guiding that the at least one second screen is moved to the display of the wearable device.
[0273] For example, the instructions, when executed by the processor, cause the wearable device to identify, based on whether the angle of the flexible display folded with the respect to the folding axis exceeds a first threshold angle, that the external electronic device is transformed into the first state. The instructions, when executed by the processor, cause the wearable device to identify, based on whether the angle of the flexible display folded with respect to the folding axis is smaller than a second threshold angle smaller than the first threshold angle, that the external electronic device is transformed into the second state.
[0274] For example, the instructions, when executed by the processor, cause the wearable device to cease, in response to transmitting a first signal to the external electronic device to display the at least one first screen through the display of the external electronic device, to display the at least one first screen on the display of the wearable device.
[0275] For example, the wearable device may comprise a second camera. The instructions, when executed by the processor, cause the wearable device to select, based on identifying a direction of gaze from an image captured by the second camera, the at least one first screen corresponding to

the direction, among a plurality of screens being displayed on the display of the wearable device.
[0276] For example, the instructions, when executed by the processor, cause the wearable device to transmit, in a state transmitting a second signal to display a plurality of screens through the display of the external electronic device, the second signal to the external electronic device, wherein the second signal includes information with respect to a layout of the plurality of screens.
[0277] For example, the instructions, when executed by the processor, cause the wearable device to display, in response to receiving a fourth signal indicating the at least one third screen from the external electronic device after transmitting a third signal to display the at least one third screen through the display of the wearable device, a visual object for guiding movement of the at least one second screen on the display of the wearable device.
[0278] For example, the instructions, when executed by the processor, cause the wearable device to identify, in a state in which the wearable device identifies the external electronic device including a flexible display insertable into a housing of the external electronic device, the state of the external electronic device as being the first state or the second state based on a size of the flexible display extracted from the housing.
[0279] For example, the instructions, when executed by the processor, cause the wearable device to identify, in a state in which the external electronic device is viewed through a preset portion within the display (**220-1**) of the wearable device, transformation of the external electronic device by using the information.
[0280] For example, the instructions, when executed by the processor, cause the wearable device to transmit, in response to identifying that the external electronic device is transformed into a first state while displaying the at least one first screen including a video, an uniform resource indicator (URI) indicating the video and a timing of the video being displayed through the at least one first screen to the external electronic device.
[0281] According to an embodiment, the non-transitory computer readable storage medium storing instructions may be provided. The instructions, when executed by an wearable device including a communication circuitry, and a display, cause the wearable device to display within the display at least one first screen. The instructions, when executed by the wearable device, cause the wearable device to transmit, in response to identifying that an external electronic device is transformed into a first state based on the information, to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device. The instructions, when executed by the wearable device, cause the wearable device to transmit, in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, information associated with at least one third screen displayed through the external electronic device. The instructions, when executed by the wearable device, cause the wearable device to display within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device on the display of the wearable device.
[0282] The apparatus described above may be implemented as a combination of hardware components, software components, and/or hardware components and software components. For example, the devices and components described in the embodiments may be implemented using one or more general purpose computers or special purpose computers such as processors, controllers, arithmetical logic unit (ALU), digital signal processor, microcomputers, field programmable gate array (FPGA), PLU (programmable logic unit), microprocessor, any other device capable of executing and responding to instructions. The processing device may perform an operating system OS and one or more software applications performed on the operating system. In addition, the processing device may access, store, manipulate, process, and generate data in response to execution of the software. For convenience of understanding, although one processing device may be described as being used, a person skilled in the art may see that the processing

device may include a plurality of processing elements and/or a plurality of types of processing elements. For example, the processing device may include a plurality of processors or one processor and one controller. In addition, other processing configurations, such as a parallel processor, are also possible.

[0283] The software may include a computer program, code, instruction, or a combination of one or more of them and configure the processing device to operate as desired or command the processing device independently or in combination. Software and/or data may be embodied in any type of machine, component, physical device, computer storage medium, or device to be interpreted by a processing device or to provide instructions or data to the processing device. The software may be distributed on a networked computer system and stored or executed in a distributed manner. Software and data may be stored in one or more computer-readable recording media.

[0284] The method according to the embodiment may be implemented in the form of program instructions that may be performed through various computer means and recorded in a computer-readable medium. In this case, the medium may continuously store a computer-executable program or temporarily store the program for execution or download. In addition, the medium may be a variety of recording means or storage means in which a single or several hardware are combined and is not limited to media directly connected to any computer system and may be distributed on the network. Examples of media may include magnetic media such as hard disks, floppy disks and magnetic tapes, optical recording media such as CD-ROMs and DVDs, magneto-optical media such as floptical disks, ROMs, RAMs, flash memories, and the like to store program instructions. Examples of other media include app stores that distribute applications, sites that supply or distribute various software, and recording media or storage media managed by servers.

[0285] Although embodiments have been described according to limited embodiments and drawings as above, various modifications and modifications are possible from the above description to those of ordinary skill in the art. For example, even if the described techniques are performed in a different order from the described method, and/or components such as the described system, structure, device, circuit, etc. are combined or combined in a different form from the described method or are substituted or substituted by other components or equivalents, appropriate results may be achieved.

[0286] Therefore, other implementations, other embodiments, and equivalents to the claims fall within the scope of the claims to be described later.

Claims

1. A wearable device, comprising: communication circuitry; a display; memory comprising one or more storage media storing instructions; and at least one processor comprising processing circuitry, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: display within the display at least one first screen; in response to identifying that an external electronic device is transformed into a first state based on the information, transmit to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device; and in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, transmit information associated with at least one third screen displayed through the external electronic device; display within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device on the display of the wearable device.

2. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: identify the external electronic

device including a hinge structure of a plurality of housings; identify the state of the external electronic device as being the first state or the second state based on an angle between housings of the external electronic device.
3. The wearable device of claim 2, further comprising a first camera, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: based on identifying the flexible display by using an image captured by the first camera, identify the angle by which the flexible display is folded with respect to the folding axis.
4. The wearable device of claim 2, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: obtain the information including the angle through the communication circuitry.
5. The wearable device of claim 2, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: while the external electronic device is transformed into the first state, display a visual object for guiding that the at least one first screen is moved to the flexible display of the external electronic device.
6. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: while the external electronic device is transformed from the second state into an intermediate state between the first state and the second state, display a visual object for guiding that the at least one second screen is moved to the display of the wearable device.
7. The wearable device of claim 2, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: based on whether the angle of the flexible display folded with the respect to the folding axis exceeds a first threshold angle, identify that the external electronic device is transformed into the first state; based on whether the angle of the flexible display folded with respect to the folding axis is smaller than a second threshold angle smaller than the first threshold angle, identify that the external electronic device is transformed into the second state.
8. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: in response to transmitting a first signal to the external electronic device to display the at least one first screen through the display of the external electronic device, cease to display the at least one first screen on the display of the wearable device.
9. The wearable device of claim 1, further comprising a second camera; wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: based on identifying a direction of gaze from an image captured by the second camera, select the at least one first screen corresponding to the direction, among a plurality of screens being displayed on the display of the wearable device.
10. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: in a state transmitting a second signal to display a plurality of screens through the display of the external electronic device, transmit the second signal to the external electronic device, wherein the second signal includes information with respect to a layout of the plurality of screens.
11. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: in response to receiving a fourth signal indicating the at least one third screen from the external electronic device after transmitting a third signal to display the at least one third screen through the display of the wearable device, display a visual object for guiding movement of the at least one second screen on the display of the wearable device.
12. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: In a state in which the wearable device identifies the external electronic device including a flexible display insertable into

a housing of the external electronic device, identify the state of the external electronic device as being the first state or the second state based on a size of the flexible display extracted from the housing.
13. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: in a state in which the external electronic device is viewed through a preset portion within the display of the wearable device, identify transformation of the external electronic device by using the information.
14. The wearable device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the wearable device to: in response to identifying that the external electronic device is transformed into a first state while displaying the at least one first screen including a video, transmit an uniform resource indicator (URI) indicating the video and a timing of the video being displayed through the at least one first screen to the external electronic device.
15. A non-transitory computer readable storage medium storing instructions, wherein the instructions, when executed by an wearable device including a communication circuitry, and a display, cause the wearable device to: display within the display at least one first screen; in response to identifying that an external electronic device is transformed into a first state based on the information, transmit to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device; and in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, transmit information associated with at least one third screen displayed through the external electronic device; display within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device on the display of the wearable device.
16. The non-transitory computer readable storage medium of claim 15, wherein the instructions, when executed by the wearable device, cause the wearable device to: identify the external electronic device including a hinge structure of a plurality of housings; identify the state of the external electronic device as being the first state or the second state based on an angle between housings of the external electronic device.
17. The non-transitory computer readable storage medium of claim 16, wherein the instructions, when executed by the wearable device comprising a first camera, cause the wearable device to: based on identifying the flexible display by using an image captured by the first camera, identify the angle by which the flexible display is folded with respect to the folding axis.
18. The non-transitory computer readable storage medium of claim 16, wherein the instructions, when executed by the wearable device, cause the wearable device to: obtain the information including the angle through the communication circuitry.
19. The non-transitory computer readable storage medium of claim 16, wherein the instructions, when executed by the wearable device, cause the wearable device to: while the external electronic device is transformed into the first state, display a visual object for guiding that the at least one first screen is moved to the flexible display of the external electronic device.
20. A method of a wearable device comprising communication circuitry and a display, comprising: displaying within the display at least one first screen; in response to identifying that an external electronic device is transformed into a first state based on the information, transmitting to the external electronic device information with respect to the at least one first screen to display at least one second screen associated with the at least one screen displayed on the display of the wearable device through the external electronic device; and in response to identifying that the external electronic device is transformed into a second state different from the first state based on the information, transmitting information associated with at least one third screen displayed through

the external electronic device; displaying within the display of the wearable device at least one fourth screen associated with the at least one third screen being displayed through the display of the external electronic device on the display of the wearable device.